MASTER THESIS

Interdependency of Bias, Fairness and Prediction Instability in Graph-based Recommender Systems

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Declaration of Authorship

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1.

Introduction

Recommender systems are crucial in today's age of information to alleviate information overload and present users with relevant information, while also helping sellers or providers with increased exposure. Their omnipresence on today's internet makes them more relevant than ever, as they provide users with personalised experiences and allow them to discover new content [7]. The ascent of recommender systems to importance started in the nineties with the rise of the World Wide Web as a platform for e-commerce. Recommender systems allowed for a new level of personalisation with regard to connecting users to content. The area of research in recommender systems has continuously grown and advanced since these early days, leveraging more and more data to increase prediction performance [4] and has evolved from neighbourhood-based methods to representation learning-based recommendation frameworks [23]. With the ever-growing amount of readily available information on the World Wide Web, recommender systems aim to mitigate the issue of information overload, be it in e-commerce settings, the news or video streaming. For recommender algorithms to be effective in their recommendations, they need to be able to accurately model user preferences [23] as their main goal is to connect users to content that is truly relevant to them [8]. They are driven by user interactions with web content such as clicks, ratings or purchases and additional information input such as user and content characteristics.

Even though recommender systems are an essential part of today's highly information-driven world, they still suffer from a number of challenges [7]. They have unparalleled influence on the type of information that users get connected to. While they offer easy access to information, their recommendation decisions are profoundly controlled by the underlying data, which in real-world scenarios is usually infused with biases [9]. These underlying biases can lead to unfair distribution of resources and information among users and make it challenging to get unbiased exposure to information [9]. Which brings us to one of the ongoing challenges in the area of recommender systems, biases in recommendation algorithms and the underlying data. Biases in recommender systems can lead to unfair recommendations and dissatisfied users, as well as providers. This challenge of balancing fairness in recommendation between different stakeholders has lead to bias and fairness becoming two of the most discussed topics in recommender systems research in

recent years [7]. This thesis focuses on the evaluation of item biases, such as recommendation diversity and popularity bias, as well as on the evaluation of user fairness. Recommendation diversity introduces the concept that, in addition to prediction performance, recommender systems should also focus on the level of heterogeneity of content in their recommendations to users [26]. Popularity bias describes the notion that recommender systems tend to focus on items that are very popular among users for suggestions, while less popular items receive fewer recommendations [7]. As recommendation algorithms serve a wide variety of users with different backgrounds and interests, we expect the algorithms to not discriminate against certain user groups and provide the similar levels of prediction performance for all users [15]. This concept is called user fairness. Even though item biases and user fairness are much discussed topics in recommender systems research, the relationships between them have not been thoroughly explored yet.

Another topic whose importance should not be underestimated is prediction instability. While there has been an increase in research regarding prediction instability in machine learning, there have been few studies focusing specifically on graph-based recommender systems [14]. Klabunde and Lemmerich [14] demonstrate that Graph Neural Networks (GNNs), a graph-based subclass of Neural Networks (NNs), exhibit considerable instability of prediction results between multiple model instantiations while still providing stable levels of prediction performance. Prediction stability ensures the reliability and reproducibility of recommendation results, which makes it an essential requirement in the recommendation context, especially in critical domains such as medicine or finance [14]. To counteract the lack of research in the aforementioned areas, the following research questions (RQs) will be evaluated during this thesis:

RQ1 How are prediction performance, item biases and user fairness influenced by diversity- and fairness-oriented training objectives?

RQ2 How do these training objectives influence the prediction instability of the respective models and the variance of recommendation objectives?

RQ3 Are there any correlations between prediction performance, item biases and user fairness?

In addition to these RQs we will also conduct different case studies evaluating the notion of user fairness proposed by one recommender algorithm and evaluating the influence of model-specific parameters on prediction results.

1.1. Contribution

Since the application of graph-based algorithms, especially GNN-based algorithms, in the area of recommender systems is fairly recent, most research conducted in the area of interdependency of item biases and user fairness has not taken graph-based recommender systems into account.

This thesis aims to give a first insight on what possible interdependencies might exist using two exemplary recommender algorithms for evaluation. Using recommender models which enable the optimisation on recommendation diversity as well as on user fairness allows us to evaluate how different focuses during model training influence the different recommendation objectives and their relationships with each other. We are additionally evaluating the connection of the different recommendation objectives to prediction instability of recommender models to uncover possible influences. While there are a few works researching relationships between individual combinations of our researched recommendation objectives in recommender systems [1; 12; 13; 17; 24], to the best of our knowledge, this is the first work to take an in-depth look into the interplay of all these factors.

1.2. Thesis structure

We will first give an overview of the foundational knowledge about recommender systems and the recent migration to graph-based recommendation, as well as the main objectives in the area of recommendation. We also introduce the evaluation measures used to evaluate the different recommendation objectives (see Chapter 2). Chapter 3 introduces the datasets and recommendation algorithms used for our experiments and describes the experimental setup for each RQ. The following Chapter presents the results obtained for the individual research questions and gives a summary and interpretation of the findings (see Chapter 4). To allow the categorisation of the results, chapter 5 introduces a selection of related research works focussing on different aspects of our overall research approach. Chapter 6 includes a conclusion of our results and the limitations we have encountered during evaluating the research questions, before giving a short outlook on possible future research topics extending our research focus.

2.

Background

The following Chapter aims to provide necessary background knowledge about recommender systems as a whole, graph-based recommendation algorithms specifically and the different recommendation objectives which will be of importance for our experiments. While biases and fairness in graph-based recommendation have been in the focus of researchers for a while, their interdependencies as well as their relationships with prediction instability are still not well researched. For a more in-depth understanding of the overall research background, the following sections provide further term definitions.

2.1. Recommender systems

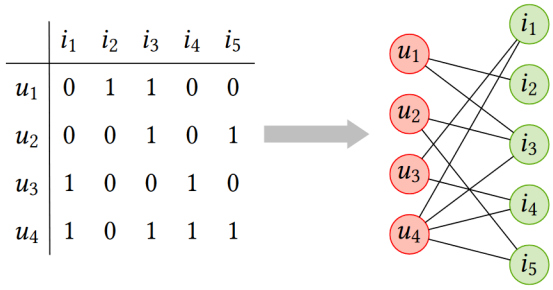
With the ever-growing amount of readily available information on the World Wide Web, recommender systems aim to mitigate the issue of information overload by connecting users to content that is relevant to them [8] through accurately modelling user preferences [23]. Due to their omnipresence and influential position as the connection between users and content, recommender systems posses exceptional influence on what content users get connected to [9]. They are primarily driven by user interactions with web content such as clicks, ratings, or purchases. In addition to the plain interaction data, some recommenders might also include additional side information like user features or content characteristics into their decision-making process [4]. We can distinguish between two different types of interaction input that recommender systems may base their decisions on, implicit and explicit feedback. Using the context of e-commerce as an example, clicks and purchases represent *implicit feedback* about a user's attitude towards certain content, which can be represented by binary rating information. *Explicit feedback* can be found where users actively rate content to show their attitude towards it. Ratings are typically given on a scale, for example between 1 and 5, that allows further specification of a user's attitude towards specific content, than just the distinction between interested vs not interested [4]. Recommender systems assume that there are certain dependencies between users and their interaction behaviour. For example, a recommendation algorithm may base its predictions for a certain user on existing inter-

action data of other users which have shown similar interaction behaviour to the original user [4]. Another possibility is for recommendation algorithms to base their predictions on past interests that a user has shown. For example, if a user has shown interest in movies of a certain genre, either by an implicit interaction or a positive rating, the recommender algorithm will be more likely to predict other movies belonging to the same or a closely related genre [4]. In this thesis, we focus on datasets and recommendation algorithms that exclusively make use of explicit user feedback as a basis for recommendation decisions and use no additional user or content characteristics. Before further specifying the recommendation context, this thesis is located in, we introduce a few basic recommendation notations: Recommender algorithms work with interaction data containing a set of unique users $\mathcal{U}$ and a set of unique items $\mathcal{I}$. Past interactions between individual users $u \in \mathcal{U}$ and individual items $i \in \mathcal{I}$ are used for training the recommendation algorithm. Based on these learned interactions, the recommendation model aims to predict the top-k most relevant items to all users in the form of a recommendation list R^k.

There are two general modes of recommendation models, a rating-based and a ranking-based mode [4]. *Learning-to-rate* recommender models make use of previously collected explicit rating data to accurately predict ratings for previously unobserved interactions between users and items, while *learning-to-rank* models focus on determining the right recommendation order for users [4]. While learning-to-rate models have been the gold standard for a long time, recent years have seen a shift to the learning-to-rank mode [13]. Similarly to the learning-to-rank mode, the learning-to-rate mode also allows for rank-based evaluations of recommendation objectives, based on continuous predicted rating scores. This thesis focuses on learning-to-rate recommender systems. This focus has been set, considering the availability of recommender models, as well as publicly available datasets fitting the research prerequisites. The learning-to-rate mode is also called the matrix completion problem. Interactions between users and items can be comprehensibly presented using a user-item matrix of dimensions $|\mathcal{U}| \times |\mathcal{I}|$, which contains all available rating information for every user $u \in \mathcal{U}$ and item $i \in \mathcal{I}$. Learning-to-rate recommendation algorithms aim to predict ratings for the remaining not-witnessed interactions between users and items [4]. A further distinction in recommender systems needs to be made between *user-item* and *user-user* recommendation. While user-item recommendation can be found, for example, in e-commerce or on news websites, user-user recommender systems are used in social networks. The research conducted in this thesis focuses on the user-item recommendation context.

2.2. Graph-based recommendation

Graph-based algorithms in the area of recommendation have been on the rise in the last few years due to their promising results regarding prediction performance, and they experience ongoing research and development [20]. Graph-based recommendation models are characterised by the graph-structured input of interaction information and possible secondary input data. Structuring input data in the form of graphs is highly useful in the area of recommendation, as it closely mir-



	i_1	i_2	i_3	i_4	i_5
u_1	0	1	1	0	0
u_2	0	0	1	0	1
u_3	1	0	0	1	0
u_4	1	0	1	1	1

Figure 2.1.: ***User-item interaction matrix and bipartite graph.*** *Figure taken from [23]*

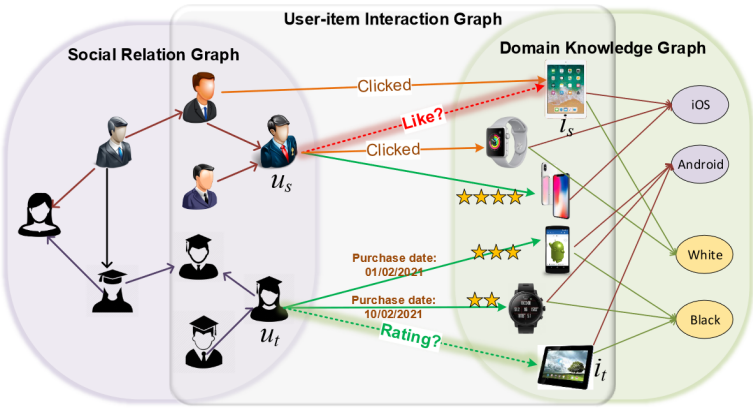


Figure 2.2.: ***Graph learning-based recommender system.*** *Figure taken from [20]*

rors the structure of the real world, where individual objects are all connected via relationships [20]. This thesis focuses on user-item recommendation, where interactions are typically represented using an interaction matrix, as shown in Figure 2.1. The content of this matrix can consist of explicit as well as implicit interaction data between users and items and can also easily be displayed using a graph structure (also shown in Figure 2.1) offering a concise representation of the relationships between different users, items and secondary information like characteristics.

In the case of a user-item interaction matrix containing scaled ratings, the corresponding graph architecture is a weighted, bipartite graph. From this graph perspective, the recommendation task at hand is a link-prediction problem on the bipartite graph [23] which translates to the learning-to-rate recommendation mode. The graph G represents the interaction data, or any other related data, through nodes N and edges E where nodes represent individual data instances, for example users and items, while edges mark a relationship between two different instances of a bipartite graph. The bipartite graph can be denoted as $G = (N, E)$ where $N = \mathcal{U} \cup \mathcal{I}$ and $E \subset (\mathcal{U} \times \mathcal{I})$. Graph representation of data does not consider each individual information input as an isolated input, but uses the network structure of the graph to exploit potential relationships [21].

Figure 2.2 shows an exemplary ecosystem of a graph learning-based recommender algorithm. The example includes a user-item interaction graph containing implicit as well as explicit interaction data, but also contains additional information input in the form of a social relation graph and a

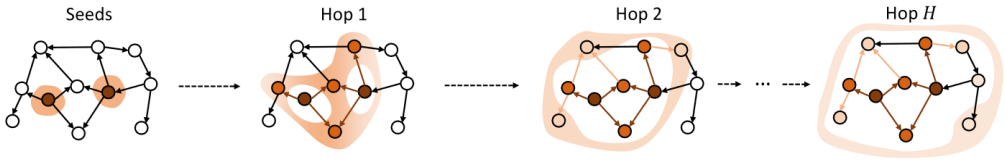


Figure 2.3.: ***Propagation of information along graph edges over multiple hops.*** *Figure taken from [19]*

knowledge graph. The advantages of graph-based recommendation models become especially evident due to their ability to incorporate such additional information in the representation learning process [23]. Transforming the recommendation method from a matrix completion approach to a graph-based approach enables the information propagation along the edges of the graph using its network structure. This mitigates classical recommendation-related challenges like data sparsity- and cold-start-related issues [19]. Data sparsity describes the challenge of training data not containing sufficient interaction data between its elements for appropriate interaction-based prediction calculations. Similarly, the cold-start problem describes the problem of calculating predictions for previously unknown elements with no interaction history [19]. The possibility for propagation of information along edges allows for the enrichment of user and item representations by considering the neighbouring instances in a graph structure (see Figure 2.3). Using graphs as a foundation for relationship-learning, allows for exploration of multi-hop relationships between nodes [23].

The most popular and recent graph-based recommendation technique are Graph Neural Networks (GNNs). GNNs are a subclass of neural networks that process data in the form of graphs [8]. GNNs have shown themselves to be superior in graph representation learning. They combine the dominant technology of Deep Neural Networks (DNNs) with graph learning-based methods to optimally leverage the graph structure of the underlying interaction data [23]. Figure 2.4 shows an exemplary structure of a GNN. For each node of the input graph, the GNN aggregates the information of its adjoining nodes. Depending on the specified neighbourhood size (number of hops), this process is performed multiple times, where each layer of the GNN is fed the aggregated graph output of its predecessor, leading to the propagation of node information over multiple hops. GNNs are able to overcome challenges prevalent in earlier stages of recommender system development, such as limited capacity and ability to observe high-order structural information in training data [8]. What makes GNNs so useful in the recommendation context, is their ability to capture relationships among users, items, and features. They can iteratively propagate and aggregate information and relations over multiple hops, thanks to multi-layer embedding propagation and have become the state-of-the-art in recommendation, achieving outstanding performances [8].

2.3. Objectives in recommendation

While traditionally the main objective of recommender systems has been to provide the highest possible accuracy regarding rating predictions (in this thesis further referenced as *prediction per-*

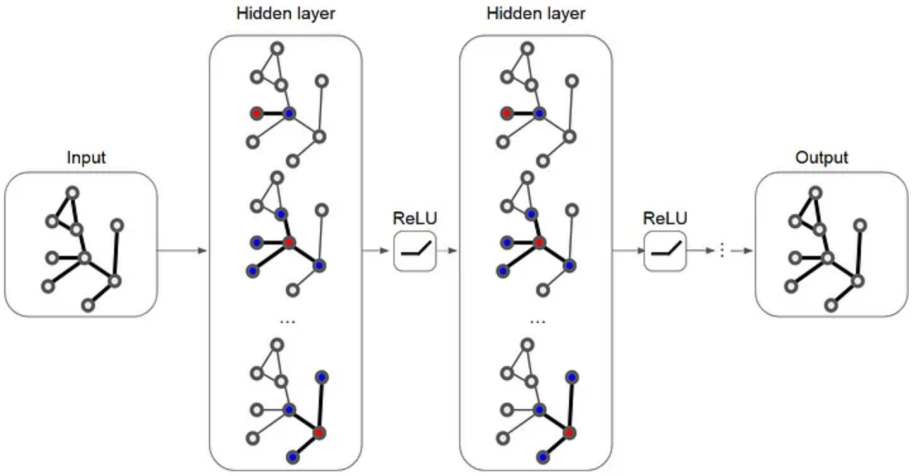


Figure 2.4.: ***Structure of a Graph Neural Network.*** *Figure taken from [18]*

formance) for unseen test items, this strict focus has loosened in recent years. The shift from the learning-to-rate to the learning-to-rank approach, aiming at giving users recommendations ranked on their perceived interest, introduced additional non-accuracy related objectives [13]. Being provided a with a limited list of top recommendation allowed for new measures to evaluate user recommendations [13]. The fast-moving research in the area of recommender systems has led to prediction performance no longer representing the sole standard for determining recommendation quality [26]. These newly added objectives are also considered challenges in the area of recommendation, especially when aiming for multi-objective-oriented recommender systems.

We can generally distinguish between three different methods to decrease biases and unfairness in machine learning [16]. First, we have the possibility of **pre-processing** training data to remove possible data-inherent biases and provide recommender algorithms with a balanced and fair data input. Alternatively, reducing biases during the training of recommender algorithms is called **in-processing** and is the most used approach when aiming for increased fairness in recommendation [16]. Finally, we also have the possibility to apply **post-processing** methods on recommendation results to remove biases through re-ranking or re-computing [16]. This thesis focuses on the in-processing approach by using recommendation algorithms that aim to debias recommendation results during application. The additional recommendation objectives, that we have focused on in our RQs, will be introduced in the following sections.

2.3.1. Prediction performance

The definition of prediction performance is dependent on the recommendation mode used by a recommender model. Due to the year-long prevalence of the learning-to-rate approach [13], most studies regarding recommender systems focus on prediction accuracy as the main evaluation focus for performance evaluation [27]. Prediction accuracy describes how accurately a recommender

system is able to predict ratings for previously unseen test ratings. As we are working with learning-to-rate recommender systems, the accuracy of predictions is measured using the ground truth rating information of test samples and the ratings predicted by the recommender algorithms. We are using the *Root Mean Square Error* (*RMSE*) to evaluate the prediction performance [6]. The *RMSE* score is calculated based on a set of evaluation samples $\mathcal{S}_e$, where each sample $s \in \mathcal{S}_e$ consists of the ground truth rating information r of a specific user u for a specific item i. The predicted rating for the user-item combination associated with a sample $s \in \mathcal{S}_e$ is defined by y_s, with r_s representing the ground truth rating. The *RMSE* score is defined as:

$$RMSE(\mathcal{S}_e) = \sqrt{\frac{\sum_{s \in \mathcal{S}_e} (\mathrm{y}_s - \mathrm{r}_s)^2}{|\mathcal{S}_e|}}. \tag{2.1}$$

2.3.2. Diversity/ Coverage

Diversity and prediction performance have a relationship going back far longer than recommender systems even exist. From being studied in the field of economics research in the 1950s to the analysis of their relationship in the area of information retrieval [13]. Diversity has evolved to a much researched topic in the area of recommender systems due to the growing consensus that a certain level of diversity in recommendation results leads to a bigger increase in user satisfaction than solely focussing on the accuracy of predictions [13]. A lack of diversity in results increases the risk of none of the recommended items fitting a user's interests and boring the users [4]. Diversifying recommendations has multi-sided advantages, for users as well as for content providers. Diversification improves the discovery potential and as a result the satisfaction for users and enhances the item visibility, especially for items from smaller providers [26].

There are different levels at which diversity can be measured in recommender systems. **User-level diversity**, also called individual diversity, describes the diversity of items in a single user's recommendation list. It is widely adopted in recommender systems as the dissimilarity of items calculated based on an item distance function [13]. **System-level diversity**, often also called item coverage, describes the overall diversification of item exposure over all users' recommendations [13]. As the terms *aggregated diversity* and *item coverage* are often used interchangeably in literature to describe the same thing, this thesis will henceforth only use the term *aggregated diversity*. Increasing diversity in recommendations often leads to a decrease in prediction performance, called the accuracy-diversity tradeoff [11].

Isufi et al. [11] propose two diversity measures, calculating the aggregated as well as individual diversity of recommendations. The aggregated diversity $AD@k$ measures the catalogue coverage of recommendations as the fraction of items $\mathcal{I}$ included in the union of all users' top-k recommendation lists, represented by R^k. The first k recommendations for a specific user $u \in \mathcal{U}$ are defined as ${R^k}_u$.

$$AD@k(R^k) = \frac{|R^k|}{|\mathcal{I}|}. \tag{2.2}$$

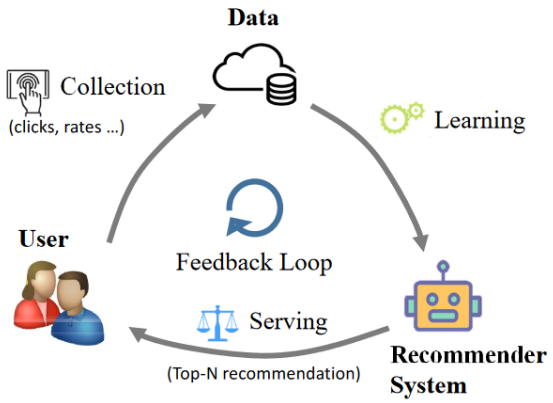


Figure 2.5.: ***Feedback Loop.*** *Figure taken from [7]*

The individual diversity $ID@k$ of recommendations is calculated using a distance metric $d(i, j)$ between two items i and j.

$$\mathrm{ID}@k(R^k) = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{2}{k(k-1)} \sum_{i \in R^k{}_u} \sum_{j \in R^k{}_u} d(i, j). \tag{2.3}$$

2.3.3. Popularity bias

Popularity bias describes the phenomenon, that recommendation systems tend to favour more popular items over unpopular items for their recommendation predictions. Popularity bias, also called the Long-tail phenomenon, originates from the uneven distribution of user ratings over items in the training data. While a small percentage of the items makes up most ratings, the remaining share of items have been rated far less frequently. Recommender systems trained on such imbalanced data tend to assign higher importance to the more popular items, thus recommending them more often than their popularity would warrant. This popularity bias leads to decreased levels of personalisation and decreased diversity of results [7].

There are two sources of item bias in recommender systems. On one side, popularity bias may originate in the input data used to train recommender models. In general, most real-world datasets suffer from a certain level of popularity bias [17]. An algorithm trained on such biased data learns to apply the biased view on data during application [15]. On the other side, recommender models may suffer from algorithmic bias, further propagating biases already existing in the input data. Recommender systems that suffer from this algorithmic bias may further intensify the level of popularity bias by recommending popular items more often than their popularity originally warrants. This concept is called Feedback Loop (see Figure 2.5). Popular items iteratively gain even more popularity, while unpopular items get less popular [17].

To evaluate the level of popularity bias, we are calculating the *Average Recommendation Popularity* ($ARP@k$) as proposed by Abdollahpouri et al. [3]. $ARP@k$ calculates an item's popularity score based on the number of ratings the item received in the training data. The $ARP@k$ score can

take on a value from $0 < ARP@k \leqslant 1$, where a value of 1 would indicate that every user is only recommended the most rated item from the training data. Using $\phi(i)$ as the number of times an item i has been rated in the training set and $\phi(i_{max})$ as the rating count of the most rated item in the training data, $ARP@k$ is calculated as:

$$ARP@k(R) = \frac{1}{\phi(i_{max})} \frac{\frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \sum_{i \in R_u} \phi(i)}{|R_u|}. \tag{2.4}$$

2.3.4. User fairness

User fairness in the recommendation context has seen an increase in attention, focusing on defining fairness as well as ensuring it in the recommendation context [26]. Similarly to popularity bias, unfairness in recommender systems, or machine learning as a whole, can generally be traced back to biases either in the underlying data or the algorithm itself [15]. For example, imbalanced user data with regard to the distribution of user ages leads to the recommendation algorithm learning a biased distribution and possibly further reinforcing it. User fairness can be broadly classified into three categories: For the evaluation of **Group fairness**, users get divided into different subgroups, usually based on their sensitive features. Such sensitive features could for example be: gender, age or ethnicity. Group fairness requires algorithms to treat protected groups (often minorities or discriminated user groups) similarly to the advantaged groups [15]. Another category of user fairness is the **individual fairness**. Individual fairness requires that similar users receive the same quality of treatment by recommender systems. Again, the similarity of users could be defined by their sensitive features. While group fairness focuses on fairness between groups consisting of similar users, individual fairness focuses on fairness in individual groups consisting of similar users [15]. The third fairness category is **hybrid fairness**, which aims to achieve multiple fairness requirements simultaneously [15]. Similarly to the accuracy-diversity tradeoff, ensuring user fairness usually also comes at the expense of prediction performance [26]. With the development of recommender systems towards graph-based recommendation, there has been research indicating that the utilisation of graph data could have a negative impact on fairness concerns, making it more challenging to counteract unfairness in recommendation [8]. There are also studies showing that optimising recommendation algorithms on diversity may have a negative influence on the level of user fairness produced by the algorithms [22]. One of the greatest challenges in the area of fairness evaluation in recommender systems is the lack of consensus regarding fairness definitions, due to varying fairness demands in different recommendation contexts. Some fairness definitions might even be inherently conflicting and cannot be realised simultaneously [15].

This thesis focuses on user group unfairness in the context of recommendation. When it comes to evaluating user group unfairness, Abdollahpouri et al. [2] define a recommender system as user unfair if it delivers varying degrees of recommendation performance across different user groups. These user groups can be based on any sensitive user attribute, for example: gender or age. We are measuring the difference of prediction performance across user groups by calculating

the coefficient of variation (CV) over the *RMSE* values of all user groups. The calculation of the CV score uses the standard deviation (σ) of a set of observations $\mathcal{T}$, as shown in Equation 2.5, where $\overline{\mathcal{T}}$ is defined as the average of the set $\mathcal{T}$.

$$\sigma(\mathcal{T}) = \sqrt{\frac{\sum_{t \in \mathcal{T}} (t - \overline{\mathcal{T}})^2}{|\mathcal{T}|}}. \tag{2.5}$$

The coefficient of variation shows the standard deviation as the ratio of the mean value to display the frequency distribution of values (see Equation 2.6).

$$CV(\mathcal{T}) = \frac{\sigma(\mathcal{T})}{\overline{\mathcal{T}}}. \tag{2.6}$$

First, we calculate the *RMSE* scores of all user groups separately (see Equation 2.7). $\mathcal{S}_G \subset \mathcal{S}$ denotes a subset of all evaluation samples belonging to user group G. Based on the *RMSE* values of all user subgroups $G \subset \mathcal{U}$, we can calculate the standard deviation σ, where $\mathcal{T}$ is a set containing the previously calculated *RMSE* values (see equation 2.5). Finally, the CV_{RMSE} score is calculated as the standard deviation over all *RMSE* scores divided by the mean of all *RMSE* scores $\in \mathcal{T}$ (see Equation 2.6).

$$RMSE(\mathcal{S}_G) = \sqrt{\frac{\sum_{s \in \mathcal{S}_G} (\mathrm{y}_s - \mathrm{r}_s)^2}{|\mathcal{S}_G|}}. \tag{2.7}$$

For an additional measure of user group unfairness, we use the *Value Unfairness* (VU) score proposed by Yao and Huang [25] as an orientation, which measures the user fairness performance of a recommender system as the inconsistency in signed estimation error across two user groups. We have adapted the approach to fit a multi-user group situation by, again, calculating the coefficient of variation over the individual VU scores of all user groups $G \subset \mathcal{U}$. In contrast to the *RMSE* score, the VU score is not calculated based on evaluation samples but uses an item subset $I_s \subset \mathcal{I}$ for calculation. $y_{(i,G)}$ is the average, predicted rating for item i over all users $u \in G$ and $r_{(i,G)}$ is the average ground truth rating for item i over all users $u \in G$. Calculating the standard deviation (see Equation 2.5) and CV_{VU} score (see Equation 2.6), $\mathcal{T}$ denotes a set containing the VU scores of all user groups.

$$VU(G) = \frac{1}{|I_s|} \sum_{i \in I_s} \left| \left(\mathrm{y}_{(i,G)} - \mathrm{r}_{(i,G)} \right) \right|. \tag{2.8}$$

2.3.5. Prediction instability

Model stability is defined as a machine learning model being stable if its performance is independent of the data used for training, as long as all model parameter settings remain the same. Prediction stability further reduces the variance in external impacts. This thesis uses the research of Klabunde and Lemmerich [14] as an orientation regarding the definition of prediction stability. A model will be characterised as stable if its performance is unaffected by using different training

seeds for model training, while all other parameter settings and the data split remain the same. Low prediction instability means low variance in rating predictions. The lower the prediction instability, the more reliable and reproducible the results [14]. The *Prediction Disagreement* (*PD*) as proposed by Klabunde and Lemmerich [14] is defined as:

$$PD(\mathcal{M}) = \frac{\sum_{m_1 \in \mathcal{M}} \sum_{m_2 \in \mathcal{M}} \sum_{i \in \mathcal{I}} \mathbb{1}\{m_1(i) \neq m_2(i)\}}{|\mathcal{M}| \cdot |\mathcal{I}|}. \tag{2.9}$$

where $\mathcal{M}$ is a set containing models that are different instantiations of the same recommendation model using identical training data and parameter settings but different training seeds. $\mathbb{1}$ is an indicator function calculating the disagreement between rating predictions of two instantiations. For every item $i \in I$, the indicator function increases its sum by 1, if the rating predictions of models *m1* and *m2* are not equal. Due to the recommender algorithms used in this thesis predicting continuous ratings for user-item interactions, we use rounded prediction values for calculating the *PD* scores.

2.3.6. Further objectives

In addition to the objectives introduced above, there are also further objectives that can be examined in the context of recommender systems.

Serendipity describes the element of surprise a user might experience being recommended an item they did not expect. Due to the challenging nature of measuring "the element of surprise", there are few formal definitions of evaluation measures. [13].

Novelty Related to the notion of serendipity, novelty in recommendations describes the fact that recommended items have been previously unknown to a user. Novelty is closely related to serendipity, as there are often overlaps in their definitions in literature. An item being unknown to a user tends to go along with the item being different from content the user has previously interacted with, thus potentially carrying an element of surprise when being recommended [13]. Like serendipity, there is no one-size-fits-all definition for novelty, as the concept of novelty allows for speculation. Recommender systems are often trained on ratings provided by users. And while it might seem obvious to define items that have not been rated by a user as novel to them, we also have to consider that users rarely rate every single item that is known to them [13].

3.

Experimental setup

The focus of this thesis is to evaluate the interdependency of bias, fairness, and prediction instability in graph-based recommender systems. The following sections introduce the datasets and recommender algorithms we have used for our experiments, and give an overview of the varying training and evaluation setups used for analysing our RQs.

3.1. Choice of datasets

Due to the characteristics and focus of our research, there are a few prerequisites, datasets need to satisfy to be suitable for usage in our experiments. As we are working with learning-to-rate recommender algorithms, the datasets need to include explicit feedback. While *implicit feedback* gives binary user interaction information, *explicit feedback* contains user interactions as scaled ratings. Candidate datasets also need to include sensitive user information, in addition to the basic rating information, for user fairness evaluation purposes. As we want to conduct our research on more than one dataset, all datasets need to include the same type of sensitive user information.

Due to the limited number of publicly available datasets satisfying our requirements, we restricted our dataset selection to the MovieLens 1M (ML1M) [10] dataset, the MovieLens 100K (ML100K) [10] dataset and the BookCrossing (BX) [28] dataset. The statistics of the datasets are shown in table 3.1. All three datasets contain additional information about user ages, which is the sensitive user feature for which we will evaluate the user fairness. Age group definitions are based on the already existing definitions used by the ML1M dataset (see Table 3.2).

	topic	**users**	**items**	**ratings**	**range**	**density**	**released**
ML100K	Movie Ratings	943	1,682	100,000	1-5	0,0630	1998
ML1M	Movie ratings	6,040	3,706	1,000,209	1-5	0,0446	2003
BX	Book ratings	3,704	130,026	246,798	1-10	0.0005	2005

Table 3.1.: *Dataset characteristics after applying pre-processing steps*

age category	0	1	18	25	35	45	50	56
ages	N/A	<18	18-24	25-34	35-44	45-49	50-55	$\geqslant 56$

Table 3.2.: *Age categories*

The BX dataset had to be pre-processed to fit the application context of this thesis. As the dataset consists of explicit as well as implicit interactions between users and items, we have removed all implicit ratings. To further downsample the number of users and items in the dataset, we have used the ML1M and ML100K dataset characteristics as an orientation and only included ratings by users that have rated at least 20 items. In contrast to the ML1M and ML100K dataset, the BX dataset still consists of a very high number of unique items which pushes the limits of the similarity calculations needed for evaluating the individual diversity of the recommendation results, which is why the BX dataset is not evaluated on this evaluation measure. We are using a data split of 80% for model training, 10% for validation and 10% for testing. All model instantiations use identical data for training, validation, and testing for better comparability and to allow the evaluation of the models' prediction instability. We have already introduced the concept of data-inherent bias as a reason for biased and unfair recommendation results in Section 2.3. To get a first impression of the biased distributions present in the introduced datasets, we do a first evaluation of the long-tail distribution of items, responsible for popularity bias, and the imbalance in distribution of age groups, responsible for user unfairness. Figure 3.1 shows the distribution of user ages in each dataset. Disregarding the users who have no associated age information (age category "0" in Figure 3.1c), we can see a very similar distribution of user ages across the different age categories. Users between 25 and 34 are the single biggest user group, closely followed by users between the ages of 35 and 44 and users between 18 and 24, in the case of the ML100K and ML1M datasets. Imbalances regarding the distribution of user groups might influence the level of user fairness produced by recommender systems.

Figure 3.2 shows the long-tail distribution of items for the datasets. All three datasets show a pronounced long-tail distribution, evident of the inherent popularity bias in each dataset. Due to the sheer number of items, combined with the high level of sparsity, the BX dataset shows a stronger long-tail distribution than the two other datasets. Based on these first evaluations, we can expect a certain level of popularity bias and user unfairness produced by the used recommender models. Figure A.1 in the appendix shows that users also more often provide higher than lower ratings for items, showing an additional imbalance of data which might lead to biased learning for recommender algorithms. Additionally, Figure A.2 shows that popular items almost exclusively receive higher averaged ratings, while unpopular items show more bandwidth with regard to average rating values. The more pronounced bandwidth might be due to many unpopular items only receiving one rating. The fact that popular items receive high averaged ratings might influence the rating level learned by the recommender algorithms,

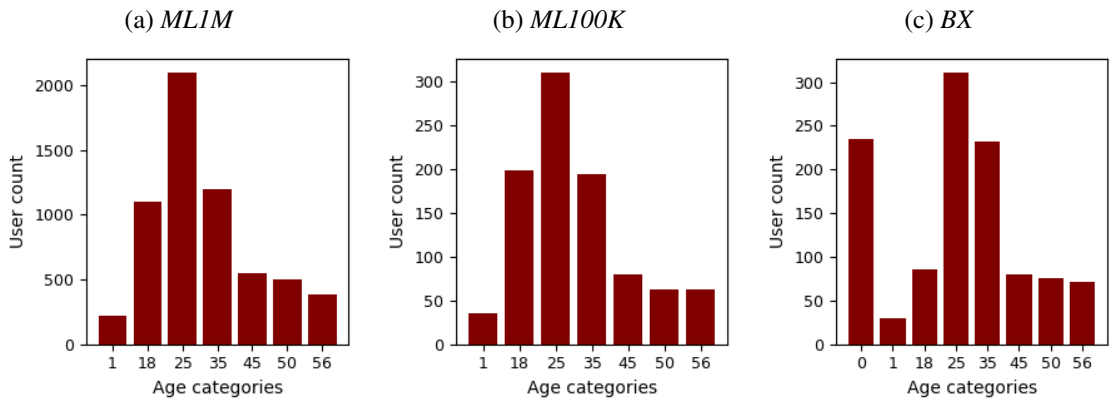


Figure 3.1.: ***Distribution of user ages over the different age categories.*** *Each category's label is the lowest age value belonging the respective category. Age category "0" for the BookCrossing dataset represents users without age information.*

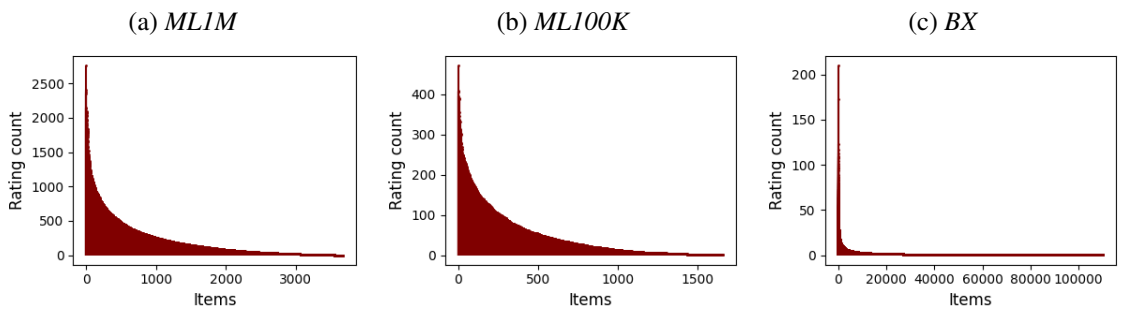


Figure 3.2.: ***Long-tail distribution of item ratings in the training data***

3.2. Recommender Algorithms

This thesis considers two different GNN-based recommender models that focus on reducing different types of bias in recommendation results. These two models have been chosen due to a lack of alternatives w.r.t. non-prediction performance-focused graph-based recommenders that operate in the context of learning-to-rate on user-item interactions. The first algorithm we will be using is the Diversity Tradeoff model as introduced by Isufi et al. [11], which focuses on the *Accuracy-Diversity Trade-off in Recommender Systems via Graph Convolutions*. In addition to prediction performance, Isufi et al. [11] also focus on individual and aggregated diversity in recommendation results. The second recommendation algorithm evaluated in this thesis is the Compositional Fairness model by Bose and Hamilton [5] focusing on *Compositional Fairness Constraints for Graph Embeddings*. The model can be optimised on prediction performance as well as on user group fairness.

3.2.1. Diversity Tradeoff model

The Diversity Tradeoff model by Isufi et al. [11] uses Graph Convolutional Filters (GCF) as a basis for a Graph Convolutional Neural Network (GCNN). The original aim of the authors is for the model to allow a tradeoff that is as advantageous as possible for prediction performance and aggregated diversity as well as individual diversity [11]. To avoid the recommendation algorithm not giving importance to items differing from the main trend and to increase diversity in recommendation results, Isufi et al. [11] consider a Nearest Neighbour (NN) graph as well as a Furthest Neighbour (FN) graph during training. Considering an exemplary node in an interaction graph, the NNs to this node can be calculated based on a given similarity metric. The goal is to gather other nodes that show the highest similarity to the original node, based on certain characteristics. The Diversity Tradeoff model calculates the similarity of nodes (either between users or items) based on the interaction matrix created from the training data. Similarly, FNs are defined as nodes that show the least similarity to the original node according to the used similarity metric. While considering only NNs for model training is associated with a higher level of prediction performance, FNs are used to aid in increasing diversity and avert the focus from mainly trend items [11]. The Diversity Tradeoff model allows to customise the extent of influence, the NN and FN graph have during training. The α parameter controls the level of influence of the FN graph. Considering the total level of influence by both graphs as 1 (100%), the α parameter can take on values $0 < \alpha < 1$, leaving the remaining influence to the NN graph (see Figure 3.3). In accordance with the original paper, we considered steps of 0.1 when increasing the α parameter.

The Diversity Tradeoff model offers a high level of customisation and exertion of influence during training. It allows the usage of both user- and item-centered NN and FN graphs, as well as the definition of the amount of considered elements when calculating the graphs. Adding a few additional parameters offering different settings, we had to reduce the amount of tested parameter combinations during parameter optimisation due to limited resources. The original paper offers a

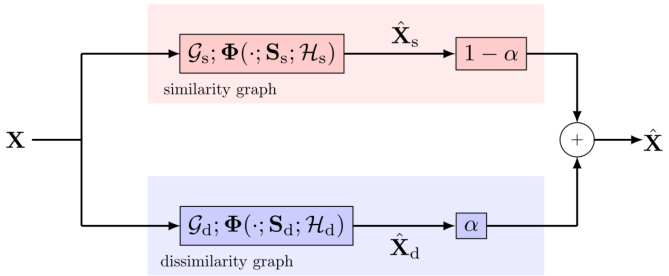


Figure 3.3.: ***Rating predictions by the Diversity Tradeoff model using similar and dissimilar graphs.*** *Figure taken from [11]*

few case studies regarding parameter settings which we have used as an orientation while downsizing the parameter pool. While the authors of the model focus on the tradeoff between prediction performance and aggregated as well as individual diversity, we focused on optimising the model on one recommendation objective at a time. Unfortunately, the model brings limitations regarding the characteristics of the underlying dataset. The Diversity Tradeoff model relies on multiple similarity and dissimilarity calculations, posing a challenge for bigger datasets. In contrast to the ML1M and ML100K dataset, the BX dataset consists of a very high number of unique items, which pushes the limits of the Diversity Tradeoff model and makes it too resource intensive to be run.

3.2.2. Compositional Fairness Model

Bose and Hamilton [5] use a Graph Convolutional Matrix Completion (GCMC) algorithm for recommendation. GCMC models combine Graph Convolutional Networks (GCNs), a subtype of GNNs, with a matrix completion approach to predict missing values for unobserved user-item interactions. The authors have used the Compositional Fairness model to increase the user fairness of recommendations while maintaining a certain level of prediction performance. Our application of the model focuses on maximising one of the recommendation objectives at a time. The definition of user fairness by Bose and Hamilton [5] is a complete invariance or independence of results with regard to the selected user feature or composition of multiple user features like age and gender. As a first step to removing the sensitive user information, the model uses a filter function for every selected sensitive feature contained in the underlying dataset. This can be done for one user feature (e.g: age) at a time or for multiple user features (e.g.: age, gender, occupation). These filter functions are applied to the embedding of every individual user. In the case of multiple filters, the model composes a filtered user embedding based on the filtered embeddings of every individual feature filter. The process of creating invariant node embeddings is shown in Figure 3.4 and Equation 3.1 where $\mathrm{C-ENC}(u, F)$ is a compositional encoder taking a user u and a set of sensitive features F as input. $f_k(\mathrm{ENC}(u))$ is the filter function corresponding to the sensitive feature k and is applied to the embedding of user u, created by the function $\mathrm{ENC}(u)$). To encourage the model to

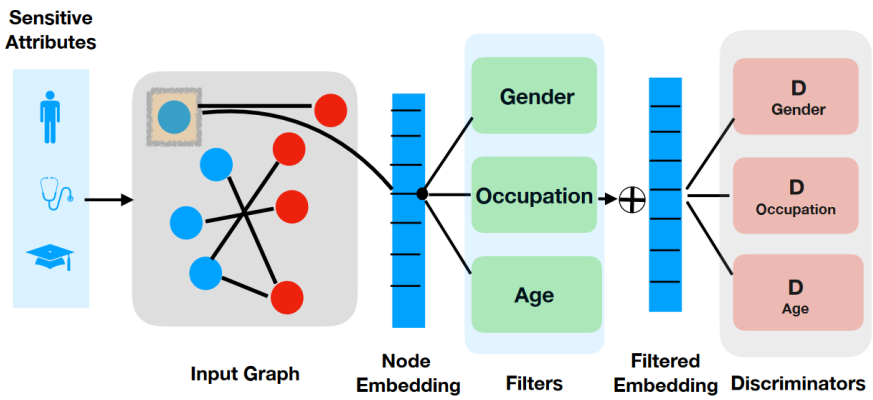


Figure 3.4.: ***Compositional Fairness model: Removing sensitive user information from user embeddings***

create invariant embeddings for different combinations of sensitive attributes during training, the input set of sensitive features F gets randomly sampled for every iteration using a binary mask.

$$\mathrm{C-ENC}(u,F) = \frac{1}{|S|}\sum_{k\in F} f_k(\mathrm{ENC}(u)) \tag{3.1}$$

To train the compositional encoder, Isufi et al. [11] make use of an adversarial regulariser that includes discriminators for every sensitive attribute k in F trying to predict sensitive features from the filtered node embeddings. The hyperparameter γ can be used to control the distinctiveness of the adversarial regularisation. Setting the value of the parameter to zero leads to no application of fairness-oriented feature filtering, while increasing the value leads to increased invariance. We have used the original paper as orientation regarding the values selected for the γ parameter, choosing a γ value of 1000 as the maximum. To evaluate the level of invariance created by the fairness filters, the authors use discriminators to predict sensitive user information based on the trained user embeddings and evaluate the accuracy of the results. In the case of the user age feature used in the experiments in this thesis, this user feature prediction accuracy is evaluated using the $F1$ score. Due to the discriminators' dependency on certain parameter settings of the Compositional Fairness model, each model instantiation trains its on discriminator. The authors show that using a compositional approach including all sensitive features for filtering user information is more effective in reducing the ability of discriminators to correctly predict user features than using filters for individual user features. Due to a lack of universally available user features in the used datasets, we will evaluate the Compositional Fairness model on its ability to only filter out the age information of users.

3.3. Training and evaluation setup

The focus of the experiments conducted in this thesis lies on evaluating the interdependency of bias, fairness, and prediction instability. To achieve that, we make use of the abilities provided by our chosen models to optimise on non-prediction performance-oriented objectives. For evaluating the different recommendation objectives, we use the evaluation measures introduced in Section 2.3. Calculating the user unfairness using the CV_{VU} measure requires a subset of items which we have fixed to a size of 1000, containing randomly chosen items for each dataset. All model instantiations use an identical selection of test items for each dataset.

We optimise the Diversity Tradeoff model [11] on: **prediction performance** using the *RMSE* evaluation measure, **individual diversity** using the *ID@20* evaluation measure and **aggregated diversity** using the *AD@20* evaluation measure. Each instantiation is trained for 50 epochs and over all instantiations the training epoch producing the best validation score for the respective recommendation objective is chosen. The Compositional Fairness model [5] is optimised on **prediction performance** using the *RMSE* evaluation measure and on **user fairness** using the *F1* score of the additional user feature classification model proposed by the authors. Each instantiation is trained for 100 epochs and over all instantiations the epoch which produces the best validation score for the respective recommendation objective is chosen for further instantiations of the model. We have run extensive parameter optimisations for every optimisation objective on each dataset using randomised training seeds. Using the combination of parameter settings which has proven itself to produce the best result for the associated training objective, we train a total of ten instantiations per optimisation objective, using identical parameter settings, only varying the training seed. Due to the resource restrictions concerning the BX dataset, we have only used is as a data basis for the optimisation objectives of the Compositional Fairness model.

RQ1 evaluates the influence of different optimisation objectives on recommendation objectives by averaging the produced evaluation scores of all ten instantiations for each evaluation measure and dataset, and additionally visualises the variance of the scores. For **RQ2**, we evaluate the prediction instability of each optimisation objective and dataset on the prediction results of all ten instantiations per optimisation objective and compare the results to the levels of variance in evaluation measures observed in RQ1. The correlation analysis conducted for **RQ3** uses all produced evaluation scores of all ten instantiations per optimisation objective and dataset. We calculate a cross-model correlation for each dataset, based on the results produced by both recommender models, and we calculate model-specific correlation coefficients for each dataset. In **case study A** we are evaluating the definition of fairness exercised by the Compositional Fairness model. For this evaluation, we use one instantiation of the model optimised on prediction performance (no fairness penalty applied) and train additional instances of the Compositional Fairness model using identical parameter settings, only increasing the fairness penalty. **Case study B** analyses the influence of the fairness penalty for the Compositional Fairness model on the different recommendation objectives evaluated in this thesis. The case study uses one instantiation of the model which

has been optimised for prediction performance and one which has been optimised for user fairness for each dataset. Using these instantiations as a starting point, we vary the level of the fairness penalty while keeping all other parameter settings, including the training seed, identical. We conduct **case study C**, evaluating the influence of the Furthest Neighbour graph during training of the Diversity Tradeoff model, similarly. For each of the used datasets, we choose one instantiation per optimisation objective and vary the influence of the FN graph when training further instantiations, while keeping all other parameter settings identical.

With regard to the fairness penalty used in the Compositional Fairness model, we use the original authors' [5] settings as an orientation regarding the choice of incremental steps used for the γ parameter controlling the fairness penalty. With a minimum value of 0 and a maximum value of 1000, the γ values are calculated by increasing the exponent in 10^x. For optimising the Compositional Fairness model on predictions performance, the γ value is fixed at 0 and for optimising the model on user fairness, the γ value is fixed 1000. Similarly, we have chosen steps of 0.1 increments for the α parameter controlling the influence of the FN graph for the Diversity Tradeoff model based on the authors' [11] choice of values. The source code used for all experiments can be found in our GitHub repository [1]

[1] https://github.com/nadine-ma/Thesis-Interdependency-of-bias-fairness-prediction-instability

4.

Results

RQ1 - Influence of different training objectives on recommendation objectives

This chapter evaluates the influence that different optimisation objectives have on the distinctiveness of the recommendation objectives introduced in Section 2.3. This evaluation is done by averaging the produced scores of all ten instantiations per optimisation objective and dataset and by additionally visualising the variance of the scores.

4.1.1. Influence on prediction performance

As already discussed in Section 2.3, prediction performance is the most used objective when working with learning-to-rate recommender algorithms. Using the *RMSE* score to evaluate how different training objectives influence the prediction performance, we obtained the results shown in figure 4.1a. As we can see in the graph, optimising both the Diversity Tradeoff and Compositional Fairness model on prediction performance, results in the best *RMSE* scores regardless of the used dataset. The Diversity Tradeoff model is able to perform rating predictions with a *RMSE* score of 0.89 for the ML1M dataset and a *RMSE* score of 1.01 for the ML100K dataset. The Compositional Fairness model shows a slightly better performance, achieving a *RMSE* score of 0.87 on the ML1M dataset and a *RMSE* score of 0.93 on the ML100K dataset. Applying the Compositional Fairness model optimised for prediction performance to the BX dataset leads to a *RMSE* score of 1.61. While this *RMSE* score is almost double of the scores achieved by the ML1M and ML100K datasets, this is due to the dataset's rating range being on a scale from 1 to 10 instead of from 1 to 5. We can also see that using non-performance-oriented optimisation objectives leads to an overall increase in *RMSE*, showing a decrease in prediction performance. With *RMSE* scores of 2.59 and 2.41 for the ML1M and ML100K datasets respectively, the Diversity Tradeoff model optimised on aggregated diversity performs worst with regard to prediction performance. The smallest dif-

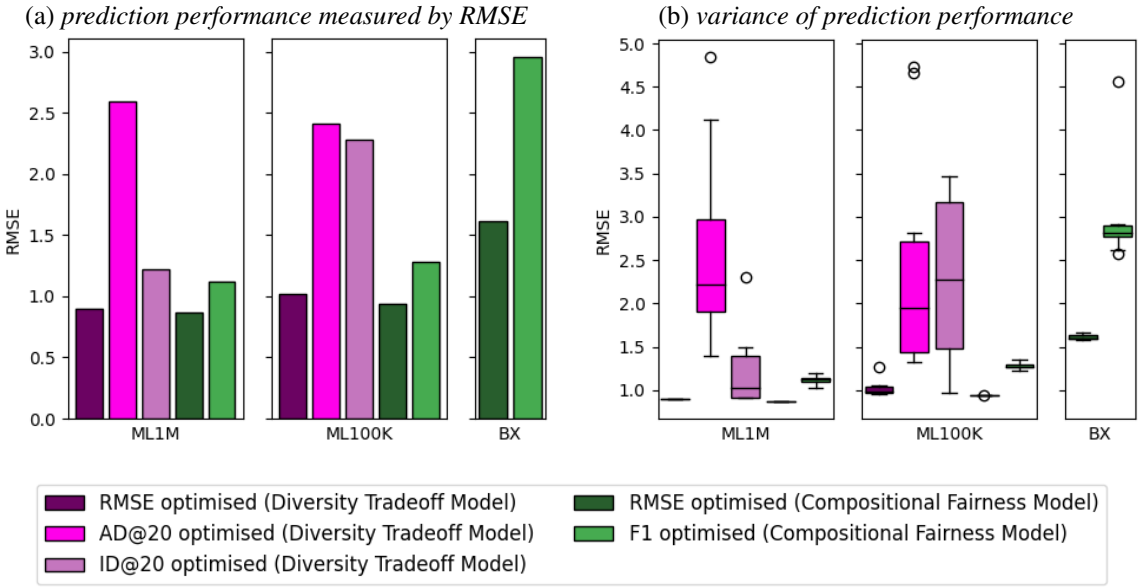


Figure 4.1.: ***Prediction performance over different optimisation objectives.*** *Optimising on non-prediction training objectives leads to less prediction performance.*

ference to the performance-optimised baseline, regarding the prediction performance, is shown by the Compositional Fairness model trained on maximising user fairness. With *RMSE* scores of 1.11 for the ML1M and 1.27 for the ML100K dataset, the model experiences far smaller performance deductions when deviating from the accuracy objective. Using the BX dataset results in a bigger difference between *RMSE* scores, as optimising for user fairness returns a *RMSE* score of 2.96. While the results shown in Figure 4.1a are of similar dimensions when varying the underlying dataset for the aforementioned training objectives, the results for optimising individual diversity show a drastic discrepancy. The *RMSE* score using the ML1M dataset is 1.22 while the score for the ML100K dataset amounts to 2.27, showing a similar dimension to the result when optimising on aggregated diversity. Except for the models trained on the aggregated diversity optimisation objective, using the ML100K dataset leads to a decrease in prediction performance in comparison to the ML1M dataset. Figure 4.1b shows the variance of results over the ten instantiations per training objective and dataset. Both models show very low variance in results when optimised on prediction performance. In contrast, we can see that the *RMSE* scores produced by the Diversity Tradeoff model, when not optimised on prediction performance, show the most variance, even when excluding outliers from the observation. Optimising the Compositional Fairness model on user fairness also increases the variance of results in comparison to the prediction performance-oriented training objectives, but on a far smaller scale compared to the Diversity Tradeoff model.

The overall finding of comparing the different training objectives w.r.t. prediction performance is that, aiming for increased diversity or user fairness in recommendation results, appears to lead to a decrease in prediction performance and to an increase in variance for the produced *RMSE* scores.

4.1.2. Influence on popularity bias

Popularity bias describes the level of difference in recommendation exposure between items of different popularity levels. A higher score of *ARP*@20 indicates a higher share of more popular items in the top@20 recommendation results for users. Using the performance-oriented training objective of both models as grounds for comparison, a first look at the plotted results in figure 4.2a shows stark differences in popularity bias. Starting with evaluating the Diversity Tradeoff model, we can see that the *ARP*@20 scores when optimising on prediction performance (0.66 for the ML1M dataset and 0.56 for the ML100K dataset) are most similar to those when optimising on individual diversity (ML1M: 0.64 and ML100K: 0.68). Applying the aggregated diversity optimisation objective during training leads to a decrease in popularity bias with *ARP*@20 scores of 0.39 and 0.37 for the ML1M and ML100K datasets respectively. The Compositional Fairness model produces overall lower levels of popularity bias than the Diversity Tradeoff model. The Compositional Fairness implementation aiming for optimisation of prediction performance returns a *ARP*@20 score of 0.30 for the ML1M dataset, a score of 0.25 for the ML100K dataset and a score of 0.013 for the BX dataset. When trained on the user fairness objective, the model achieves even lower levels of popularity bias with *ARP*@20 scores of 0.09 and 0.10 for the ML1M and ML100K dataset respectively and a score of 0.007 for the BX dataset. The fact that the BX dataset produces significantly lower levels of popularity bias, might be due to the dataset containing far more items and a higher share of unpopular items than the two MovieLens datasets.

Figure 4.2b shows the distribution of the individual *ARP*@20 data points over all ten instantiations per training objective and dataset. What is noticeable is that the Diversity Tradeoff model optimised for aggregated diversity shows a far higher range of values than all other training objectives. The lowest *ARP*@20 score achieved on the ML1M dataset is 0.02 and the highest one 0.70. The lowest variance over both datasets is shown by the Diversity Tradeoff model and Compositional Fairness model, both optimised on prediction performance.

4.1.3. Influence on aggregated diversity

As discussed in Section 2.3.2, we calculate aggregated diversity as the catalogue coverage of dataset items in all users' top@20 recommendations. Figure 4.3a shows the values for the *AD*@20 evaluation measure for the different optimisation objectives. The lowest *AD*@20 score is produced by the Compositional Fairness model optimised for prediction performance on the BX dataset, with a score of 0.005. This low score in comparison to the other two datasets can most likely be attributed to the high sparsity of the dataset, complicating the process of the recommender algorithm learning relationships between items. Due to the *AD*@20 score being calculated as the percentage of coverage of the dataset, the BX dataset containing circa 130,000 items is disadvantaged regarding aggregated diversity compared to the other datasets. This fact makes it even more surprising that optimising for user fairness on the BX dataset leads to a *AD*@20 score of 0.26. To achieve this level of aggregated diversity, the model needs to recommend on average 9.12

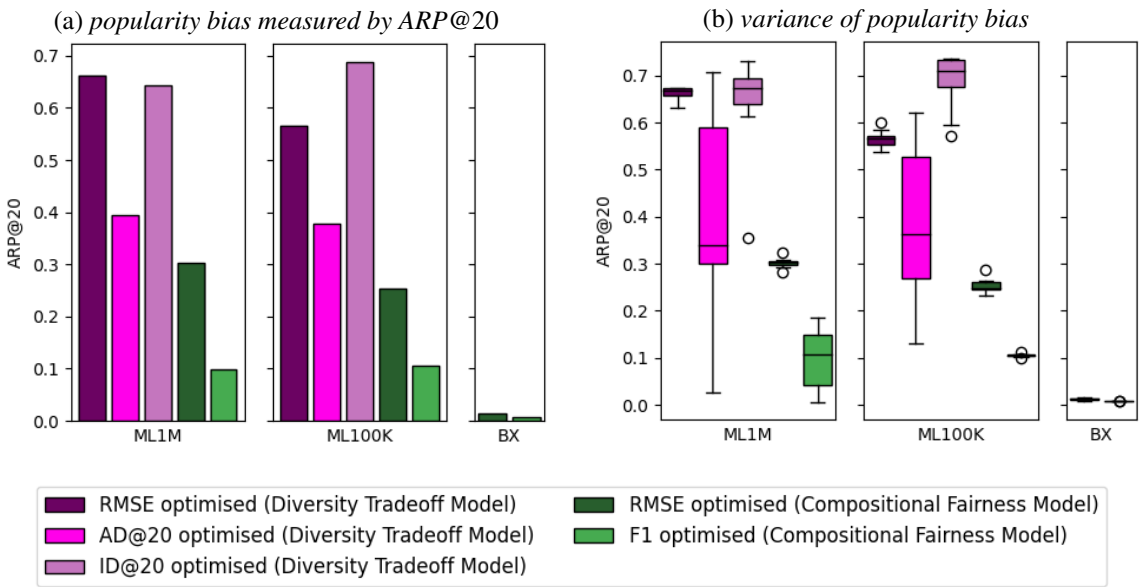


Figure 4.2.: ***Popularity bias over different optimisation objectives.*** *The Compositional Fairness model produces overall less popularity bias than the Diversity Tradeoff model.*

unique items to every user in the dataset. Comparing the non-prediction performance-focussed optimisation objectives to the performance-focussed baselines, we can see that optimising for aggregated diversity and user fairness leads to an increase in aggregated diversity while optimising for individual diversity decreases the aggregated diversity level. Optimising for individual diversity produces a $AD@20$ score of 0.25 on the ML1M dataset and a score of 0.11 on the ML100K dataset. Unexpectedly, optimising for aggregated diversity does not lead to the highest averaged $AD@20$ score when using the ML100K dataset with a score of $AD@20 = 0.67$. Applying the aggregated diversity objective to the ML1M dataset in turn produces the highest average $AD@20$ score, with a value of 0.65. The high level of variance in the results pushes down the value, as the median value amounts to a $AD@20$ score of over 0.80 (see Figure 4.3b). Applying user-fairness optimisation on the ML1M dataset leads to an average $AD@20$ score of 0.60 and on the ML100K dataset to the best score of $AD@20 = 0.93$. Comparing the individual instantiations' values for every training objective and not just the averages (4.3b), we can see that the Diversity Tradeoff model optimised for aggregated diversity on the ML1M dataset actually does provide the single highest $AD@20$ score over all instantiations with a value of 0.94. The Compositional Fairness model optimised for user fairness using the ML100K dataset, on the other hand, shows negligible variance of its high results over its instantiations. In contrast, the results for the ML1M dataset show strong variance, with the lowest value being 0.21 and the highest being 0.91. Again, the models using prediction performance-focussed training objectives show the overall least variance of $AD@20$ values.

In comparison to the prediction performance-oriented training objective, the Diversity Tradeoff model optimised for aggregated diversity increases the $AD@20$ score and the individual diversity-

(a) *aggregated diversity/coverage measured by AD@20*

(b) *variance of aggregated diversity*

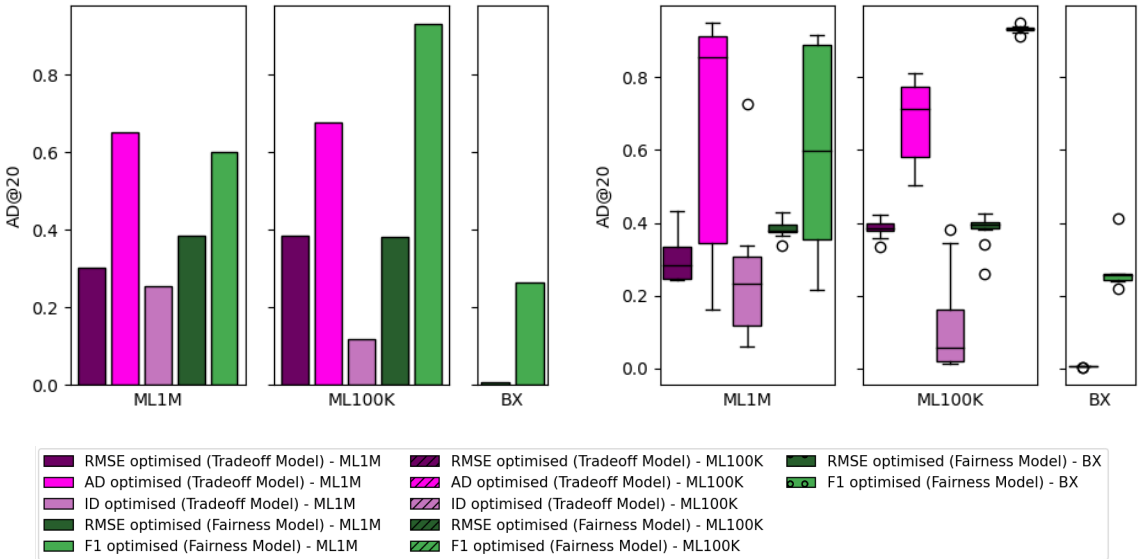


Figure 4.3.: ***Aggregated diversity over different optimisation objectives.*** *Optimising for aggregated diversity and user fairness produces the highest levels of aggregated diversity.*

oriented approach decreases the $AD@20$ score. This might be a first indicator for aggregated diversity and individual diversity showing contrasting behaviour. Optimising on user fairness also leads to an increase in aggregated diversity.

4.1.4. Influence on individual diversity

Due to the high number of items in the BX dataset requiring too many computational resources for distance calculations, we were unable to gather results regarding the $ID@20$ evaluation measure for this dataset. The first striking characteristic of the comparison graph shown in figure 4.4a is the difference between levels of individual diversity for the two datasets. Every training objective leads to higher values of $ID@20$ using the ML100K dataset. The graph also shows a difference between the two used recommendation algorithms. The Compositional Fairness model produces lower levels of individual diversity than the Diversity Tradeoff model. While the highest scores of $ID@20$ using the ML100K dataset are produced by the Diversity Tradeoff model optimised on the individual diversity objective, with a value of 0.239, this does not hold true for the ML1M dataset. Due to one low-value outsider, the individual diversity training objective produces a lower average score of $ID@20 = 0.156$ than the performance-oriented objective, with a score of $ID@20 = 0.158$. Removing the low-value outsider would lead to an average $ID@20$ score of 0.161 for the ML1M dataset. Optimising for aggregated diversity lowers the level of individual diversity in recommendation results in comparison to the prediction performance-oriented baseline of the

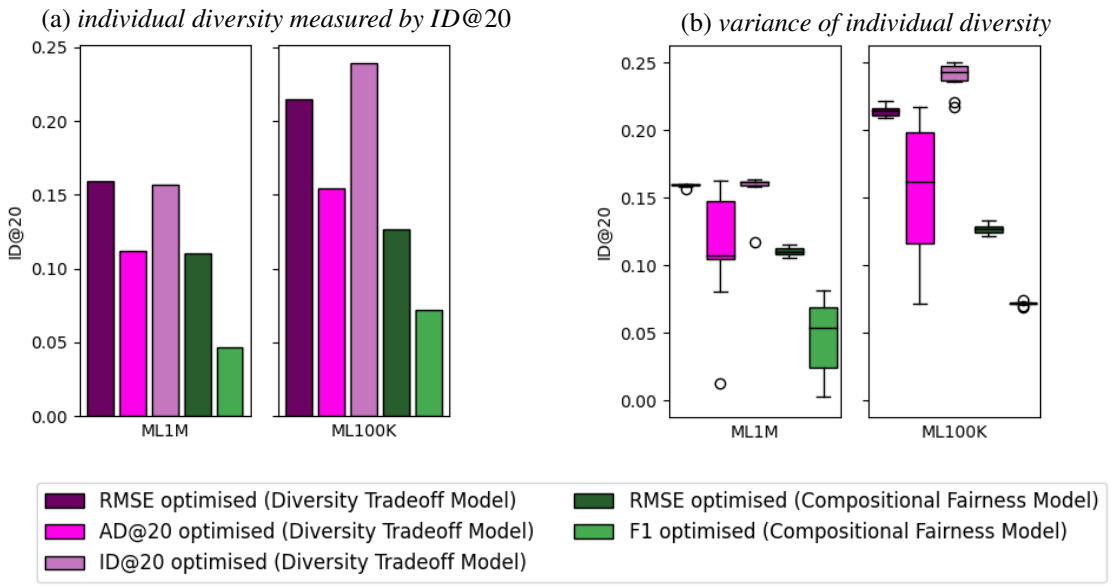


Figure 4.4.: ***Individual diversity over different optimisation objectives.*** *Optimising on prediction performance and individual diversity produces the highest levels of individual diversity.*

Diversity Tradeoff model. The same development can be seen for the Compositional Fairness model, when optimising on user fairness in comparison to prediction performance.

These averaged results imply that optimising on aggregated diversity or user fairness decreases the level of individual diversity in recommendation results. Figure 4.4b shows the variance of results for the different optimisation objectives. Optimising on aggregated diversity shows the highest level of variance, while optimising on prediction performance shows the lowest levels, regardless of the used recommender model. Optimising the recommendation results on individual diversity shows overall less variance than the other diversity-oriented training objective, but nonetheless also produces a few outliers.

4.1.5. Influence on user unfairness

CV$_{RMSE}$ Figure 4.5a shows the average CV_{RMSE} scores, as an evaluation measure of user unfairness. Using the ML1M dataset, the Diversity Tradeoff model optimised for prediction performance performs worst regarding the unfairness measure with a value of 0.10, closely followed by the Diversity Tradeoff model optimised for individual diversity with a score of 0.088. Training the model on aggregated diversity produces the best CV_{RMSE} score for the Diversity Tradeoff model using the ML1M dataset, with a value of 0.075. The results obtained using the ML100K dataset paint a different picture. Using the smaller dataset produces the worst CV_{RMSE} score for the aggregated diversity training objective with a value of 0.066 and the best score for the model optimised on individual diversity with a score of 0.030, closely followed by a score of 0.042 when optimising for prediction performance. The same behaviour can also be seen when evalu-

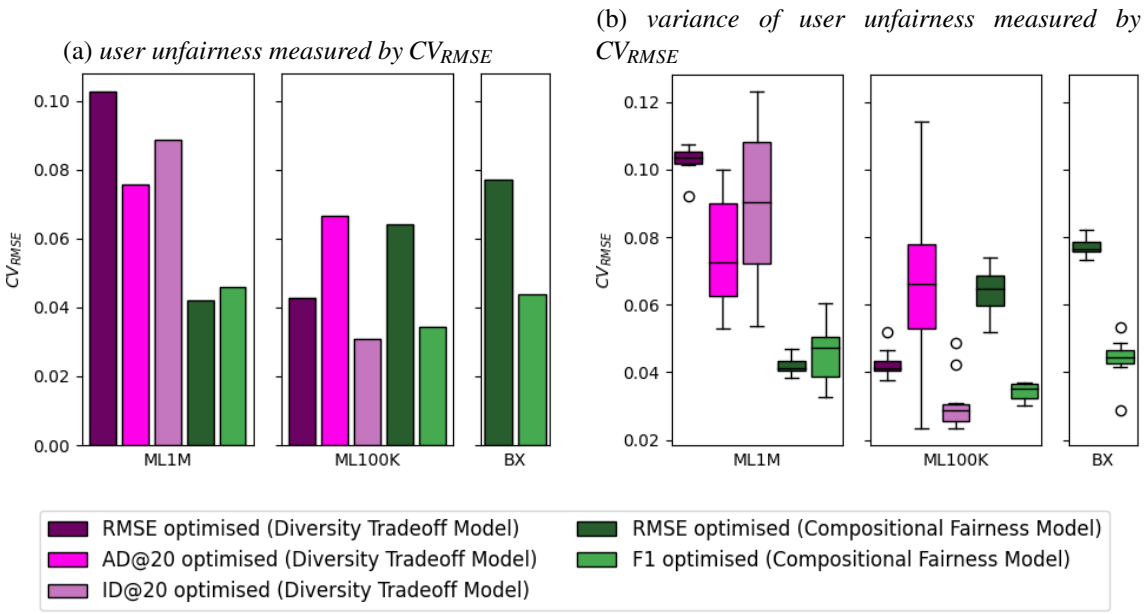


Figure 4.5.: ***User unfairness, measured by CV_{RMSE} over different optimisation objectives.*** *The graph shows varying results for the different datasets for some optimisation objectives*

ating the optimisation objectives applied to the Compositional Fairness model. While optimising for best possible prediction performance leads to the best user fairness on the ML1M dataset ($CV_{RMSE} = 0.041$) and a slightly worse score when optimising for user fairness ($CV_{RMSE} = 0.045$). Using the ML100K dataset, the model produces a better CV_{RMSE} score when training on user fairness ($CV_{RMSE} = 0.034$) and a worse score for the model optimised on prediction performance ($CV_{RMSE} = 0.063$). Similarly to the ML100K dataset, the BX dataset also shows a decrease in user unfairness measured by CV_{RMSE} when optimising the Compositional Fairness model on user fairness. While the model achieves a CV_{RMSE} score of 0.077 when training for best possible prediction performance, this score decreases to a value of 0.043 when optimising for user fairness.

Similarly to the previous graphs, figure 4.5b shows again that training the Diversity Tradeoff model on maximising aggregated diversity leads to the highest levels of variance across datasets. Optimising on individual diversity shows the second-highest level of variance, while both recommender models produce the least variant results when trained on prediction performance.

CV_{VU} Figure 4.6 shows the influence that optimising for different objectives has on the level of user unfairness measured by the CV_{VU} score. Evaluating the Diversity Tradeoff model, we can see that optimising for prediction performance provides the lowest user unfairness according to the CV_{VU} score. Using the ML1M dataset produces a CV_{VU} score of 0.026 while using the ML100K dataset leads to an even lower score of 0.019. As previously mentioned in Section 2.3.4, there are studies indicating that optimising recommendation algorithms on diversity might have a negative influence on the level of user fairness produced by the algorithms [22]. Applying the aggregated diversity optimisation objective during training increases the user unfairness to a value of 0.048 for

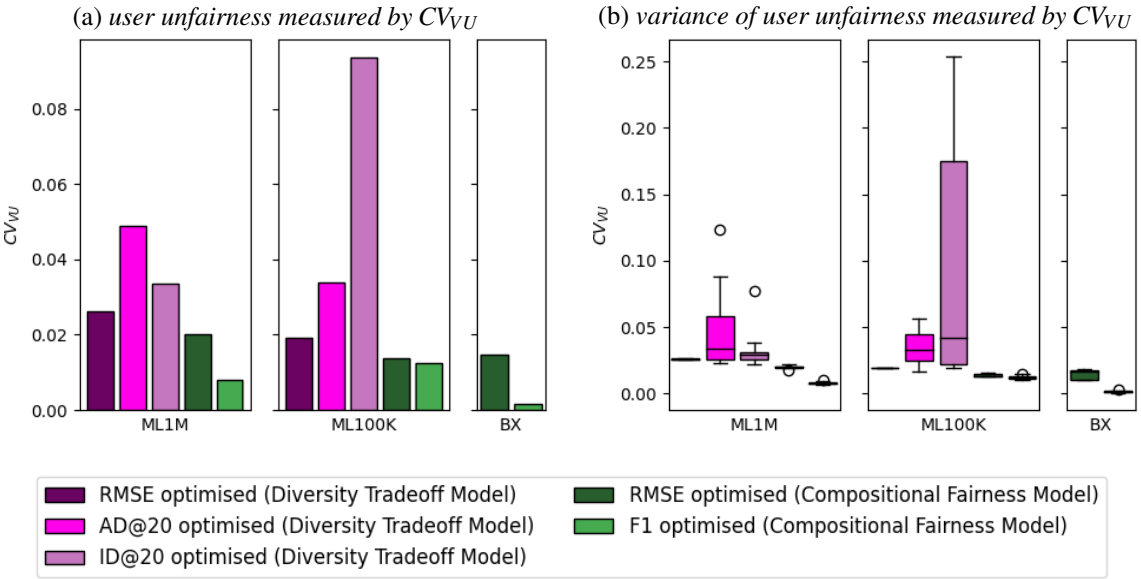


Figure 4.6.: ***User unfairness, measured by CV_{VU} over different optimisation objectives.*** *Optimising on user fairness decreases the user unfairness measured by CV_{VU} in recommendation results.*

the ML1M dataset and a value of 0.033 for the ML100K dataset. Using the ML1M dataset when training for individual diversity leads to a CV_{VU} score of 0.033, showing more user unfairness than the prediction performance-oriented objective and less unfairness than the aggregated diversity optimisation objective. The ML100K dataset, on the other hand, produces a CV_{VU} score of 0.093 which is a steep increase compared to both other Tradeoff training objectives. When looking at the corresponding variance distribution shown in figure 4.6b we can see that the reason for the steep increase is a high level of variance over the instantiations of this training objective. In comparison to the average value, the median CV_{VU} score over all instantiations amounts to circa 0.045, showing considerably less increase in user unfairness. Analysing the Compositional Fairness model, we can see that, for every dataset, applying the user fairness training objective leads to a decrease in user unfairness measured by CV_{VU}. The model produces CV_{VU} scores of 0.019, 0.013 and 0.014 for the ML1M, ML100K and BX dataset respectively when aiming for optimised prediction performance. Switching to the user fairness training objective decreases these values to 0.007 for the ML1M dataset, 0.012 for the ML100K dataset and 0.001 for the BX dataset. Looking at the levels of variance produced by each objective (see Figure 4.6b) we can see that the training objectives for the Compositional Fairness model as well as the prediction performance-oriented objective for the Diversity Tradeoff model produce the lowest levels of variance. The single highest level of variance in results is produced by the individual diversity optimisation objective applied to the ML100K dataset.

4.1.6. Summary and interpretation of results

Evaluating the overall influence of different optimisation objectives, we can conclude the following observations. In comparison to the prediction performance-focused optimisation objective, we saw a decrease in prediction performance when optimising on training objectives focused on decreasing item biases or user unfairness. Due to the data-inherent popularity bias and user imbalance, which we have already proven in Section 3.1, the recommender models striving for best prediction performance learn these biases to produce predictions close to reality. Unlearning these biases thus decreases the level of prediction accuracy. Optimising on prediction performance also showed the least levels of variance over the different model instantiations. A further evaluation w.r.t. the variance levels of evaluation measures will be done for RQ2. Focusing on maximising the individual diversity during model training, we observed a small increase in individual diversity for most model instantiations and a decrease in aggregated diversity. The level of popularity bias observed in recommendations also increased for most instantiations when optimising on individual diversity. Popular items receive their popularity status because they apply to a wider range of users. More popular items thus tend to show higher values of similarity scores calculated based on their interaction data. According to this, a simultaneous increase or decrease in individual diversity and popularity bias is not the expected behaviour of a recommendation algorithm. When optimised on individual diversity, the Diversity Tradeoff model does not seem to increase the individual diversity of user recommendations by increasing the overall number of items in all users' top-k recommendations, but by increasing the share of popular items in recommendations while ensuring that individual users receive more dissimilar recommendations. An exemplary visualisation of this recommendation setting is shown in Table 4.2. Regarding the user unfairness, the individual diversity optimisation objective showed contrasting results for the two evaluation measures used in this thesis. While the user unfairness according to the SC_{RMSE} score decreased, the user unfairness according to the CV_{VU} score increased. The decrease in the SC_{RMSE} score suggests that the Diversity Tradeoff model optimised for individual diversity, while being less accurate in predictions, increases the equality of the error level across the different user groups. The difference in behaviour between the two evaluation measures indicates, that the two definitions of user unfairness do not necessarily correlate with each other.

Optimising for aggregated diversity, showed contrasting behaviour for the item bias evaluation measures compared to the results obtained from optimising for individual diversity. While the aggregated diversity in recommendation results increased considerably, the level of individual diversity was negatively influenced by the training objective. This is an unexpected result, as we would expect the overall diversification of results to also increase the individual diversity of recommendations due to more unpopular items needing to be accommodated in recommendation results. Along with the decreased individual diversity, we also observed less popularity bias in the recommendation results. The decrease of popularity bias is expected, as the recommendation algorithms need to propose numerous unpopular items in addition to the more popular ones to achieve a high level of aggregated diversity. An exemplary recommendation representation for the rela-

tionships between popularity bias, aggregated and individual diversity is shown in Table 4.1 The level of user unfairness according to the CV_{VU} measure increased in comparison to the prediction performance-focused objective. We have not evaluated how the interest in popular items differs between the different user groups, but a possible imbalance in this area could be the reason for the increase in the CV_{VU} score. As especially the prediction quality for users interested in popular items suffers from enforcing higher aggregated diversity. We could not observe a universally valid result regarding the CV_{RMSE} user unfairness measure, which might be attributed to differences in the ML1M and ML100K dataset.

Table 4.1.: *AD* ↑, *ID* ↓, *ARP* ↓

User 1	User 2	User 3
A1	B6	C8
A4	B9	C3
A7	B2	C5

Table 4.2.: *AD* ↓, *ID* ↑, *ARP* ↑

User 1	User 2	User 3
A1	A1	A1
B2	B2	B2
C3	C3	C3

Table 4.3.: ***Exemplary recommendation results to visualise relationships between popularity bias (ARP), aggregated diversity** (AD) **and individual diversity** (ID).* *The table content displays different items, where items starting with the same letter are defined as being similar to each other. The numbering defines the popularity rank of each item, with the 1 representing the most popular item.*

Lastly, optimising for increased user fairness also led to a decrease in the prediction performance of the recommendation algorithm, but on a smaller scale than for the other non-prediction performance-oriented objectives. Regarding the item bias evaluation measures, the model optimised for user fairness showed similar changes as the aggregated diversity-oriented optimisation objective. The user fairness objective decreased the level of popularity bias and individual diversity shown in the recommendation results, and led to an increase in aggregated diversity in the results. But unlike the model optimised for aggregated diversity, the user fairness objective reduced the level of user unfairness measured by the CV_{VU} score. Regarding the CV_{RMSE} score, for two out of the three tested datasets, the model decreased the level of user unfairness. Based on these results, we can formulate the theory that the definition of user unfairness by Bose and Hamilton [5], the authors of the Compositional Fairness model, might be similar to the definition of unfairness using the CV_{VU} score.

RQ2 - Prediction instability and variance in evaluation measures

Figure 4.7 shows the level of *Prediction Disagreement* (*PD*) that has been calculated based on ten instantiations per optimisation objective and dataset, using identical parameter settings with varying training seeds. Analysing the influence of varying optimisation objectives on the Diversity Tradeoff model, we can see that aiming for best prediction performance produces the lowest levels

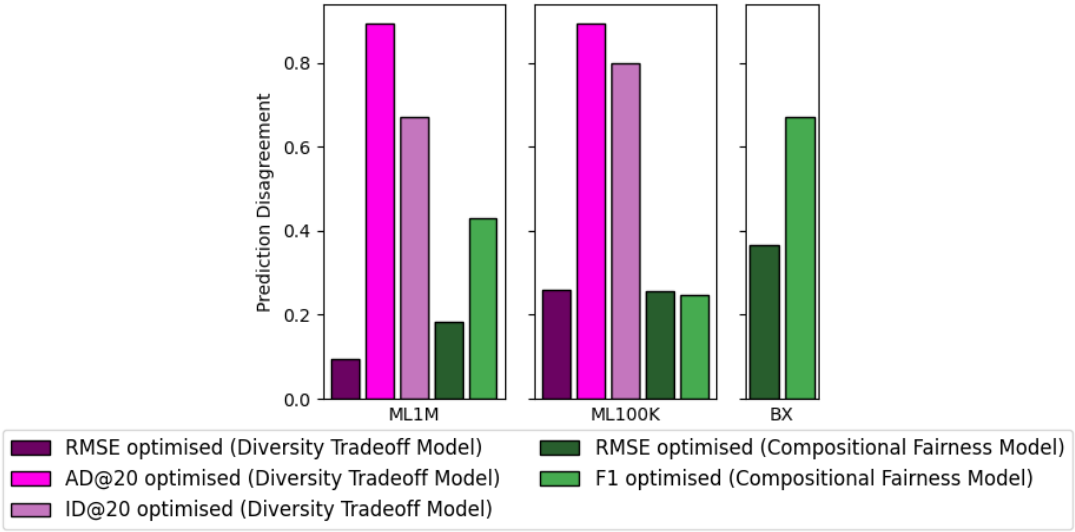


Figure 4.7.: ***Prediction Disagreement** (PD) **over different optimisation objectives** Optimising on non-prediction performance-oriented optimisation objectives leads to an increase in prediction instability measured by PD*

of prediction instability. The *PD* score achieved on the ML1M dataset amounts to 0.09 while the *PD* score for the ML100K dataset is 0.25. To illustrate the meaning of the *PD* score: Out of all different combinations of two instantiations' user rating predictions for one test item, the models will need to predict different ratings for 25% of the tested samples. Optimising on aggregated diversity of recommendation results produces a *PD* score of 0.89 for both the ML1M and ML100K dataset. One could assume that the model might have reached its highest possible prediction instability, considering that a *PD* score of 1.0 indicates that every single instantiation out of all ten instantiations in total would need to predict a different rating for every single test element. Similarly to the aggregated diversity training objective, optimising on individual diversity also leads to higher values of *PD* for both datasets ($PD = 0.67$ for the ML1M dataset and $PD = 0.79$ for the ML100K dataset). We can also see similar behaviour when evaluating the Compositional Fairness model, examining the results for the ML1M and BX datasets. Optimising on prediction performance using the Compositional Fairness model produces a *PD* score of 0.18 for the ML1M dataset and a *PD* score of 0.36 for the BX dataset. Changing the training objective to focus on user fairness instead, the prediction disagreement for these two datasets increases to 0.43 and 0.67 for the ML1M and BX dataset, respectively. A different development can be observed when using the ML100K dataset. Optimising on prediction performance leads to a *PD* score of 0.25 and optimising on user fairness leads to a slightly lower *PD* score of 0.24.

4.2.1. Summary and interpretation of results

When evaluating the influence of different optimisation objectives on the prediction instability of the respective model, the most prominent finding is that optimising on non-prediction

performance-oriented objectives typically leads to an increase in prediction instability, regardless of the used dataset. The levels of prediction instability, measured by Prediction Disagreement (see Figure 4.7), show similar characteristics to the plot displaying the influence of the different training objectives on the prediction performance (see Figure 4.1a). Optimising on aggregated diversity showed the highest level of instability, closely followed by the models optimised on individual diversity, while optimising on user fairness showed the least increase in prediction instability compared to the prediction performance-oriented baseline. One exception from this pattern can be seen for the ML100K dataset, which showed a slight decrease in prediction instability when optimised on user fairness. Comparing the prediction instability scores for the Compositional Fairness model with the prediction performance scores, we can see that the BX dataset actually provides a very similar level of prediction performance compared to the other two datasets when normalising the *RMSE* scores over the rating range of each model, but demonstrates higher prediction instability. This could be due to the higher sparsity of the dataset or the strongly imbalanced ratio between users and items in the dataset. When comparing the prediction instability results to the boxplot diagrams shown in Section 4, displaying the variances of the individual evaluation measures, we can see a very similar behaviour. Both the Diversity Tradeoff model and the Compositional Fairness model showed the lowest levels of variance in evaluation measures when optimised on prediction performance. The overall highest variance was shown by the aggregated diversity-focused optimisation objective, followed by the results for the individual diversity-focused optimisation objectives. Out of all non-performance-oriented objectives, the user fairness optimisation objective showed the overall lowest levels of variance for evaluation measures. While optimising for prediction performance often is comparably straightforward and does not leave much leeway to algorithms when it comes to producing the best possible recommendation result, especially diversification of results can be achieved by multiple alternative solutions and allows for more exploration of the overall graph space. This can lead to higher prediction instability and also seems to have a negative influence on the stability of evaluation measures.

RQ3 - Correlation analysis

After analysing the influences, different optimisation objectives have on the distinctiveness of evaluation measures using averaged values, the next step is to do a more in-depth analysis of the correlations between these evaluation measures. For the correlation analysis, we are no longer averaging the results over all ten instantiations per training objective and dataset but are using each individual instance as information input for relationship analysis. The correlation matrices visually display the strength and orientation of the correlation between two recommendation objectives, using colour blocks. We have used the Spearman correlation to evaluate the ranked correlation of the data points over all fifty runs per dataset. We are using the Spearman correlation, as the focus of the correlation analysis does not lay on analysing the linearity of relationships but on the overall ranking-based relationship between evaluation measures. A correlation value of +1/-1

represents a perfect positive/negative correlation between two recommendation objectives, while the complete absence of any correlation would return a value of zero. Correlation coefficients from 0-0.3 indicate a negligible level of correlation, and scores between 0.3-0.5 indicate a low level of correlation. A moderate correlation between two variables is shown by a correlation coefficient between 0.5–0.7. A high level of correlation can be assumed if the correlation coefficient lies above 0.7, with scores above 0.9 indicating a very high correlation. This scale is applicable to positive as well as negative correlation coefficients. A positive correlation indicates that two variables show similar behaviour regarding growth or decrease in value, while a negative correlation indicates contrasting behaviour. We are calculating the correlations between the different evaluation measures separately for every dataset and recommender model, as well as calculating a combined correlation over both models.

Figure 4.8 shows the cross-model correlations for the ML1M and ML100K dataset individually, each calculated based on fifty results, ten results per optimisation objective, thirty of which have been produced by the Diversity Tradeoff model, twenty by the Compositional Fairness model. We can see that there are similarities in the high correlation areas over both datasets. However, the datasets show contrasting behaviour w.r.t. the CV_{RMSE} score. The distinctiveness of correlations is overall lower for the ML1M dataset than for the smaller ML100K dataset. Figure 4.9 shows the correlations between evaluation measures solely based on the results from optimisation objectives for the Diversity Tradeoff model. The correlation values are calculated based on thirty results total per dataset. On a first look, we can see that the correlation matrices show differences to the cross-model correlation matrices for their respective dataset. However, what stands out, is a certain level of similarity in the high to very high correlation areas of the matrices. Lastly, figure 4.10 shows the correlation results for the Compositional Fairness model, based on twenty results per dataset. While all three correlation matrices show similarities in a few areas, the ML1M matrix shows profound differences in comparison to the ML100K and BX matrices. We will now have a closer look at the individual correlations, discussing them in descending order based on the cross-model calculations, starting with higher correlations.

***ID*@20 vs *ARP*@20** The cross-model correlation between individual diversity measured by *ID*@20 and popularity bias measured by *ARP*@20 amounts to 0.94 for the ML1M dataset and to 0.98 for the ML100K dataset (see Figure 4.8). These scores indicate a very high, positive correlation between the two evaluation measures when taking both the results from the Diversity Tradeoff model and from the Compositional Fairness model into consideration. Evaluating the relationships between the two evaluation measures separately for each model shows only small differences regarding the strength of correlation. Applying the Diversity Tradeoff model to the ML1M dataset produces a correlation of 0.83, using the ML100K dataset produces a correlation of 0.98 (see Figure 4.9). The highest correlation coefficient is achieved by the Compositional Fairness model in combination with the ML1M dataset, reaching a value of 0.99. The ML100K dataset still reaches a very high, positive correlation with a score of 0.96 when used for the Compositional Fairness optimisations (see Figure 4.10). Figure 4.11 shows scatter plots depicting the relationship between

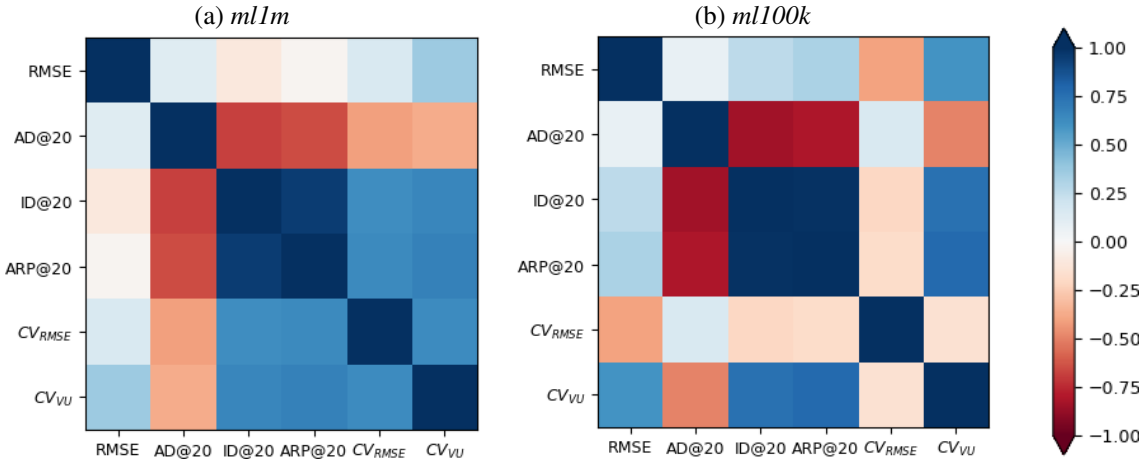


Figure 4.8.: ***Correlation over all optimisation objectives of both models.***

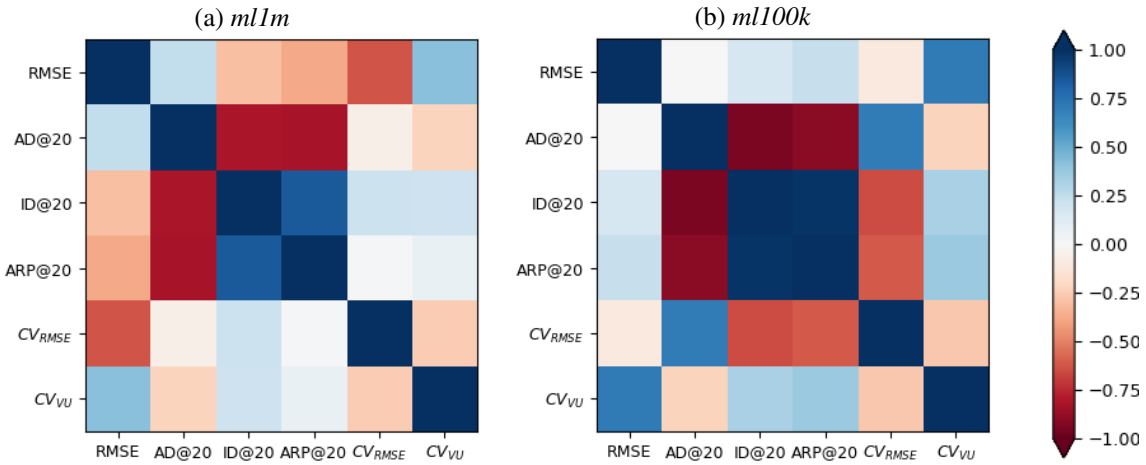


Figure 4.9.: ***Correlation over all Diversity Tradeoff model optimisation objectives.***

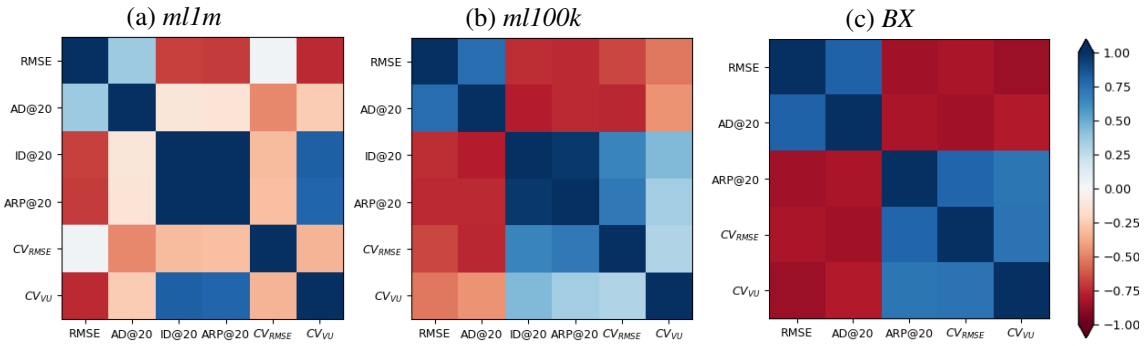


Figure 4.10.: ***Correlation over all Compositional Fairness model optimisation objectives.***

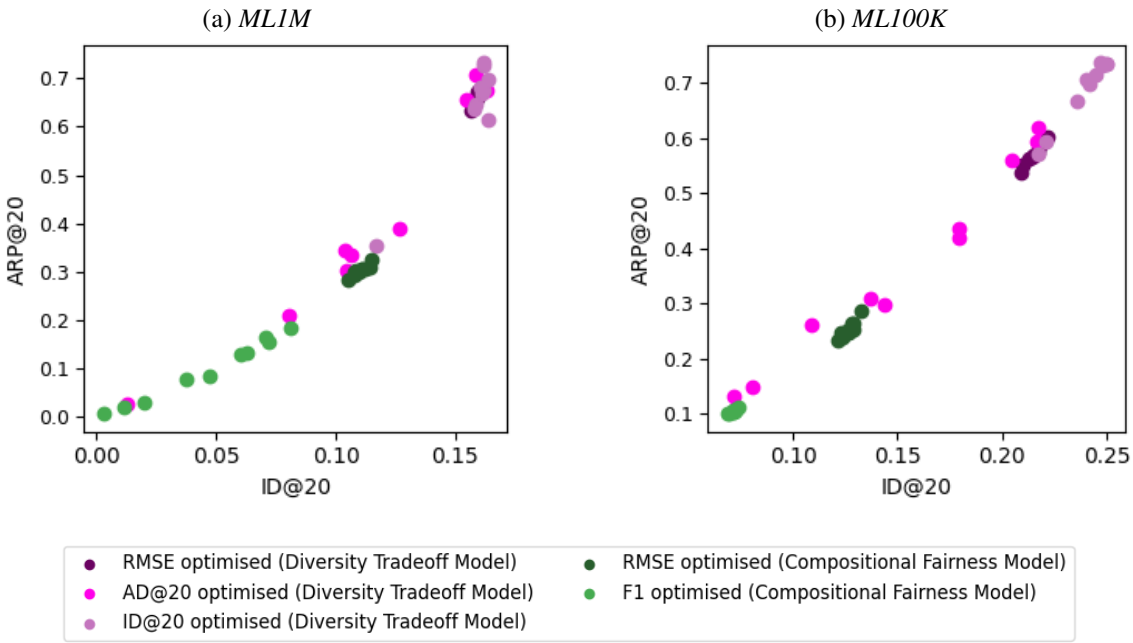


Figure 4.11.: ***Scatter plots for individual diversity and popularity bias over all optimisation objectives*** *show a pronounced positive correlation over all optimisation objectives.*

the *ID*@20 and *ARP*@20 scores produced by all optimisation objectives. As described earlier, both the ML1M and the ML100K dataset show a strong positive relation. While the relationship between the two evaluation measures is overall monotonic for both datasets, the results from the ML100K dataset also show an overall linear relationship. Due to the non-feasibility of calculating the individual diversity for the BX dataset, we cannot present results regarding that dataset. The results indicate that both recommender models and both the ML1M and the ML100K dataset show high, to very high, positive correlation between individual diversity measured by *ID*@20 and popularity bias measured by *ARP*@20. Increasing the individual diversity goes along with increased popularity bias in recommendation results and vice versa.

***AD*@20 vs *ID*@20** With one exception, the relationship between aggregated diversity and individual diversity shows an overall high, negative correlation across both models. Comparing the correlations, considering all optimisation objectives of both recommender models (see Figure 4.8), the correlation coefficient for the ML1M dataset amounts to a value of -0.68, just under the threshold for a high, negative correlation, and the coefficient for the ML100K dataset is -0.83, making it a high, negative correlation. The results obtained by just comparing the data points from the Diversity Tradeoff model (see Figure 4.9) show an even more pronounced correlation between the aggregated and individual diversity. While the training objectives applied to the ML1M dataset now reach a high, negative correlation with a coefficient of -0.81. The ML100K dataset produces a correlation coefficient of -0.94, showing a very high, negative correlation. When looking at the correlation results for the Compositional Fairness model, we can see a strong difference between

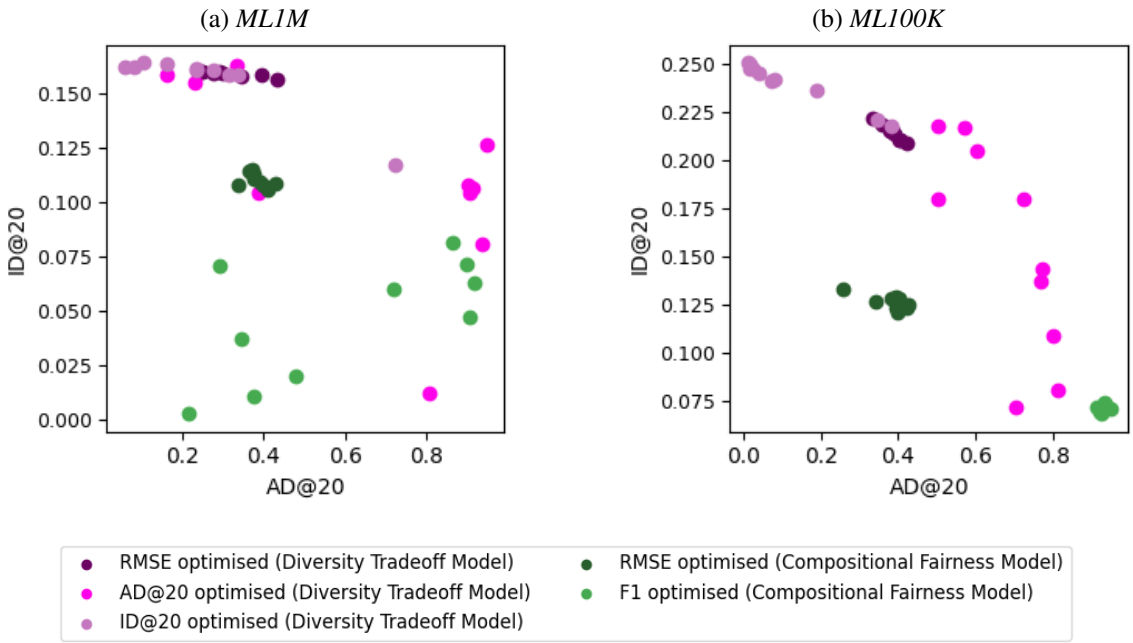


Figure 4.12.: ***Scatter plots for aggregated and individual diversity over all optimisation objectives*** *show an overall negative correlation.*

the results for the ML1M and ML100K dataset (see Figure 4.10). While applying the model to the ML100K dataset also leads to a high, negative correlation coefficient of -0.78, the ML1M only shows negligible correlation with a coefficient of -0.11. The negligible correlation for the ML1M dataset using the Compositional Fairness model might explain the lower level of correlation for the dataset, when looking at the overall correlation results of both models. The reason for the low correlation coefficient can be seen in Figure 4.12a where the individual data points obtained from optimising the Compositional Fairness model on user fairness using the ML1M dataset show very high variance in comparison to the results obtained using the ML100K dataset (see Figure 4.12b). For comparison, the scatter plots also show the overall monotonic relationship between the *AD@20* and *ID@20* scores for the Diversity Tradeoff model, where the difference in distinctiveness between the ML1M and ML100K dataset is nicely visualised. While the data points obtained using the ML100K dataset show a strongly monotonic downwards trend, the ML100M shows more variance, especially regarding the results from the aggregated diversity-oriented optimisation. The cross-model moderate to high, negative correlations as well as the mostly high correlations for individual models indicate a contrasting relationship between aggregated and individual diversity. The higher the *AD@20* score produced by a model, the lower the *ID@20* score and vice versa (except for the Compositional Fairness model applied to the ML1M dataset. The high level of variance in the results does not allow for any conclusions).

AD@20* vs *ARP@20 The relationship between aggregated diversity measured by *AD@20* and popularity bias measured by *ARP@20* closely mirrors the previously evaluated relationship be-

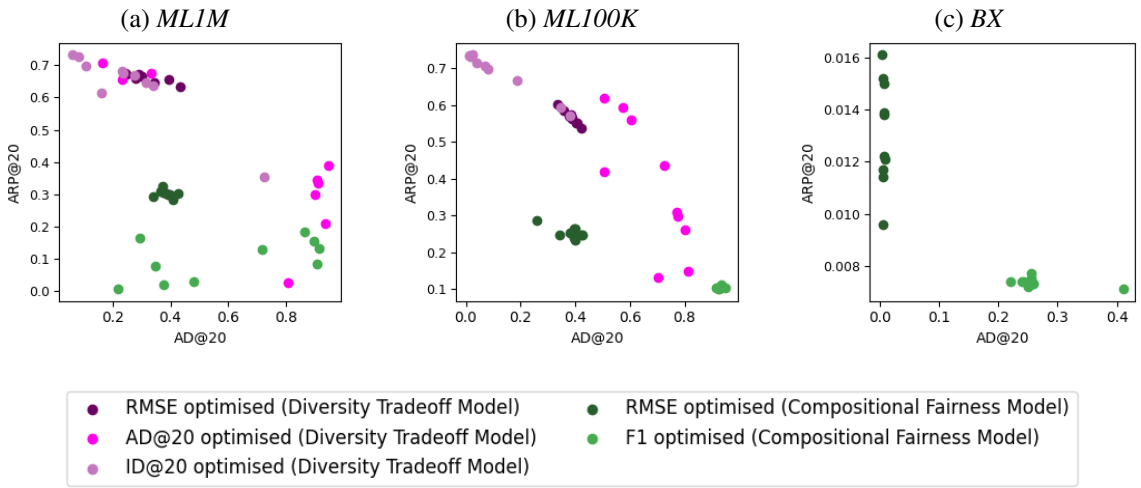


Figure 4.13.: ***Scatter plots for aggregated diversity and popularity bias over all optimisation objectives*** *show an overall negative correlation especially for the Diversity Tradeoff model.*

tween aggregated and individual diversity. Calculating the overall correlation over both the Diversity Tradeoff and Compositional Fairness model, we can see a moderate, negative correlation with a score of -0.65 for the ML1M dataset and a strong, negative correlation with a score of -0.81 for the ML100K dataset. The Diversity Tradeoff model alone produces a high, negative correlation coefficient of -0.82 for the ML1M and a very high, negative correlation coefficient of -0.90 for the ML100K dataset. Again, evaluating the Compositional Fairness model on its own shows a strong divergence between the ML1M and ML100K as well as the BX dataset. With a correlation coefficient of -0.13 for the ML1M dataset, the relationship between *AD@20* and *ARP@20* shows a similarly negligible correlation as the previously evaluated relationship between *AD@20* and *ID@20*. The reason being again the high level of variance produced by the user fairness optimisation objective, which can be seen in Figure 4.13a. The high level of variance does not allow for a distinct relationship between aggregated diversity and popularity bias. In contrast, applying the Compositional Fairness model to the ML100K and BX dataset leads to a strong, negative correlation with scores of -0.74 and -0.81 respectively. The scatter plots for both the ML1M and ML100K dataset (see Figure 4.13) show the distinct relationship between aggregated diversity and popularity bias produced by the Diversity Tradeoff model.

The overall correlation scores indicate that the recommender models produce results of contrasting behaviour regarding aggregated diversity and item popularity bias. An increase in aggregated diversity (measured by *AD*@20) goes along with a decline in popularity bias (measured by *ARP*@20) and vice versa. This result supports the statement by [7] that increased popularity bias is connected to lower aggregated diversity due to the recommender models mainly recommending a few popular items to all users, which negatively influences the overall share of recommended items.

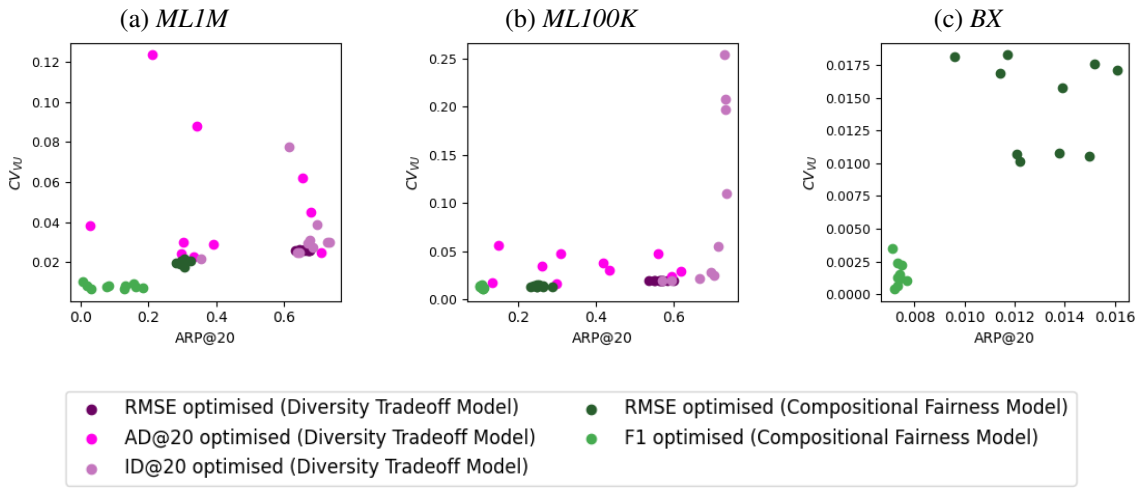


Figure 4.14.: ***Scatter plots for popularity bias and user unfairness (CV_{VU}) over all optimisation objectives.***

ARP@20* vs *CV_{VU} Applying all optimisation objectives of the Tradeoff and Compositional Fairness model to the ML1M dataset, leads to a moderate, positive correlation of 0.67 between item popularity bias measured by *ARP@20* and user fairness measured by CV_{VU}. The ML100K dataset produces a higher, positive correlation of 0.77 over both recommender models (see Figure 4.8). Analysing the relationship between the evaluation measures using the Diversity Tradeoff model (see Figure 4.9) shows an almost non-existent correlation between *ARP@20* and CV_{VU} on the ML1M dataset, producing a correlation coefficient of 0.07. With a correlation coefficient of 0.36, the ML100K dataset shows a higher, but still low, positive correlation. The highest correlation can be seen when evaluating the training objectives of the Compositional Fairness model on the ML1M dataset (see Figure 4.10a). With a high, positive correlation score of 0.89, the ML1M produces a considerably stronger relationship between the two evaluation measures than the ML100K dataset with a score of 0.34. Finally, applying the Compositional Fairness training objectives to the BX dataset leads to a correlation coefficient of 0.71 between popularity bias and user fairness, measured by CV_{VU}. Looking at the distribution of the data points for the BX dataset (see Figure 4.14c) we can see a considerable level of variance for the results obtained by the user fairness-oriented training objective. Nevertheless, the Compositional Fairness model still achieves a high, positive correlation.

The fact that the cross-model correlation for the ML1M dataset is considerably higher than the correlation evaluated only on the Tradeoff dataset, shows that the data points produced by the Compositional Fairness model strongly support the overall correlation. This can also be seen in the scatter plot shown in Figure 4.14a. The scatter plot for the ML100K dataset 4.14b shows a very visible monotonic relationship between the two evaluation measures for the Diversity Tradeoff model optimised on individual diversity. Only evaluating the correlation for the ten instantiations optimised for individual diversity produces a high, positive correlation coefficient of 0.86.

***ID@20* vs *CV*$_{VU}$** Individual diversity measured by *ID*@20 and user unfairness measured by CV_{VU} show a moderate to high, positive correlation when evaluated across both models. Using the ML1M dataset, the relationship shows a correlation coefficient of 0.65, the ML100K dataset increases this correlation to a score of 0.74 (see Figure 4.8). Evaluating only the Diversity Tradeoff model optimisation results, the ML1M dataset produces a negligible correlation score of 0.20 and the ML100K dataset produces a low, positive correlation coefficient with a value of 0.32 (see Figure 4.9). The optimisation objectives for the Compositional Fairness model return a high, positive correlation with a value of 0.81 for the ML1M dataset and a low, positive correlation with a value of 0.44 for the ML100K dataset. The scatter plot visualising the spread of data points for both the ML1M and ML100K dataset can be found in the appendix (see Figure B.3). The cross-model correlation indicates moderate to high, positive correlation between individual diversity and user unfairness, measured by CV_{VU}. An increase in popularity bias tends to go along with an increase in user unfairness.

***RMSE* vs *CV*$_{VU}$** When comparing the correlation between prediction performance measured by *RMSE* and user fairness measured by CV_{VU} over all optimisation objectives (see Figure 4.8), the ML1M dataset produces a low, positive correlation with a correlation coefficient of 0.36. The coefficient for the ML100K dataset amounts to 0.60, showing a moderate, positive correlation. When considering both recommender models individually, they show contrasting correlations between the two evaluation measures. The Diversity Tradeoff model shows a low, positive correlation with a coefficient of 0.42 for the ML1M dataset and a high, positive correlation with a coefficient of 0.70 for the ML100K dataset. With a score of -0.74 for the ML1M dataset, a score of -0.52 for the ML100K dataset and a score of -0.85 for the BX dataset, the Compositional Fairness model shows a moderate to high, negative correlation between prediction performance and user fairness measured by CV_{VU}. The difference in relationships depending on the used recommender model can be well seen in Figure 4.15a. The data points belonging to training objectives of the Diversity Tradeoff model indicate an overall more positive correlation trend, while the data points belonging to the Compositional Fairness model show a contrasting downwards-oriented trend. This downwards trend is less pronounced for the ML100K dataset, where the CV_{VU} scores produced by the fairness-optimised model instantiations show a similar level of user unfairness as the results produced by the prediction performance-oriented instantiations. Using the Compositional Fairness model, an increase in user unfairness tends to correlate with a decrease in prediction performance, while for the Diversity Tradeoff models the opposite holds true. This correlation example indicates that there is not one universally valid correlation between the two recommendation objectives.

***AD@20* vs *CV*$_{VU}$** The CV_{VU} score shows a somewhat steady correlation level with the *AD*@20 score over the ML1M and ML100K dataset. The lowest levels of correlation are shown when examining the relationship between the two evaluation measures for the Diversity Tradeoff model (see Figure 4.9). The ML1M dataset produces a negligible, negative correlation coefficient of -0.22 and the ML100K dataset, on a similar level, also shows a negligible, negative correlation

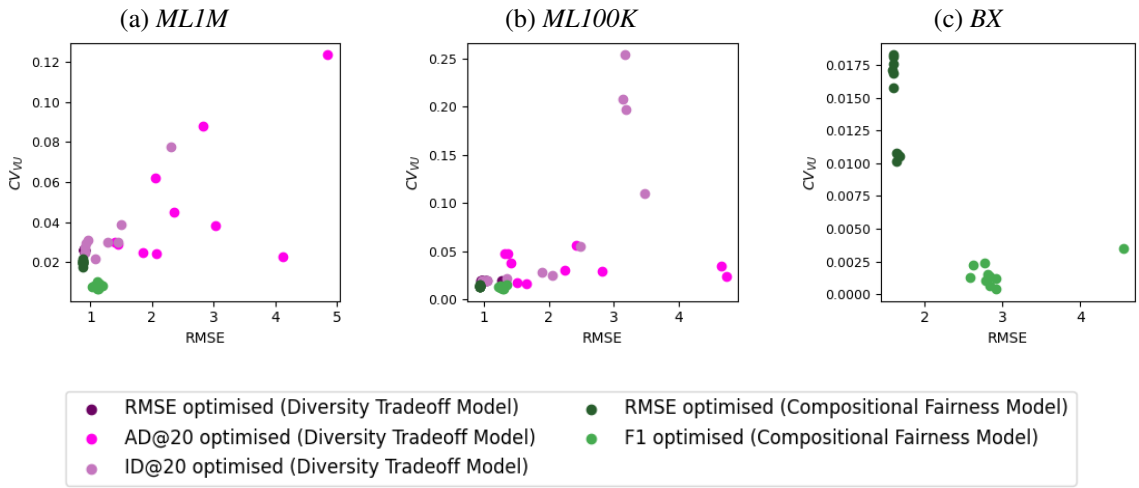


Figure 4.15.: ***Scatter plots for prediction performance and user unfairness (CV_{VU}) over all optimisation objectives.***

with a score of -0.21. Evaluating the distribution of data points in the scatter graphs (see Figure 4.16), we can see the widespread distribution of results for the Diversity Tradeoff model optimised for aggregated diversity. But we can also see the near perfect monotonic correlation for the data points obtained by optimising on individual diversity using the ML100K dataset. Running the Compositional Fairness model on the ML1M and ML100K datasets also leads to negligible or low, negative correlation (see Figure 4.10). The ML1M dataset achieves a correlation coefficient of -0.25 and the ML100K dataset a correlation coefficient of -0.45. Overall, taking both recommender models into consideration for evaluation, the two datasets produce low, negative correlations with a correlation coefficient of -0.36 for the ML1M dataset and a correlation coefficient of -0.49 for the ML100K dataset. Only the BX dataset, which is only evaluated in relation to the Compositional Fairness model, produces a higher correlation between aggregated diversity and user unfairness measured by CV_{VU}. With a score of -0.78 the two evaluation measures show a high, negative correlation. These results do not indicate a relevant level of interdependency between the two evaluation measures.

ID@20* vs *CV_{RMSE} Comparing the cross-model relationship between individual diversity measured by $ID@20$ and user unfairness measured by CV_{RMSE}, we can see that the ML1M and ML100K dataset show very different expressions of correlation (see Figure 4.8). With a correlation coefficient of 0.61 the ML1M dataset shows an overall moderate, positive correlation while the ML100K dataset shows a negligible, negative correlation with a correlation coefficient of -0.20. Only evaluating data points belonging to the Diversity Tradeoff model, we can deduce a negligible, positive correlation with a score of 0.21 for the ML1M dataset and a moderate, negative correlation with a score of -0.64 for the ML100K dataset (see Figure 4.9). The Compositional Fairness model produces a low, negative correlation with a correlation coefficient of -0.31 for the ML1M dataset and a moderate, positive correlation for the ML100K dataset, reaching a score of

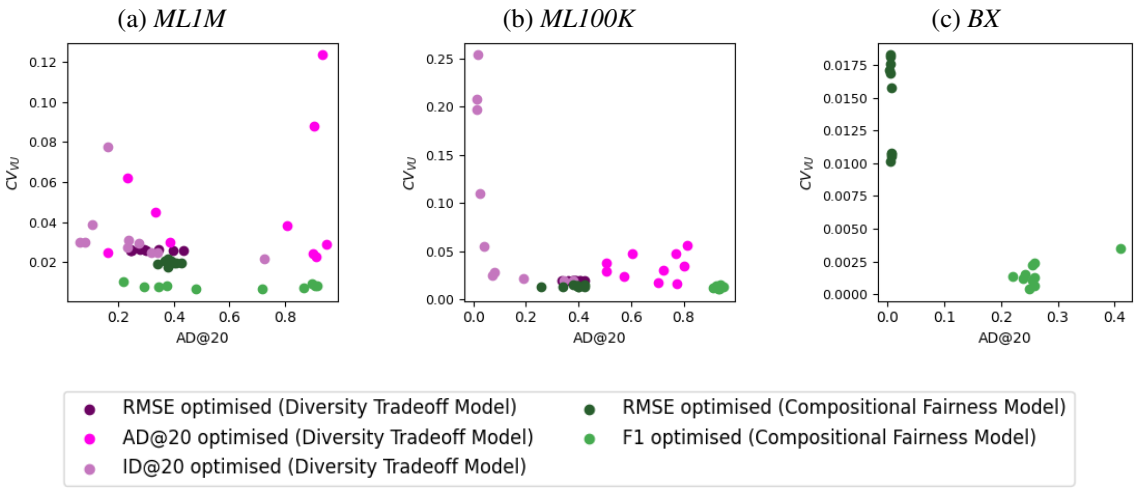


Figure 4.16.: ***Scatter plots for aggregated diversity and user unfairness (CV_{VU}) over all optimisation objectives.***

0.66. The scatter plot depicting the individual data points and their relationships can be found in the appendix (see Figure B.2). Based on these results, we cannot deduce a universally applicable relationship between the two evaluation measures, due to their distinctiveness being highly dependent on the underlying dataset.

ARP@20* vs *CV_{RMSE} The strength of the correlation between popularity bias measured by *ARP@20* and user unfairness measured by CV_{RMSE} is also highly dependent on the model providing the results and the underlying dataset. The overall correlation between the two evaluation measures amounts to 0.63, a moderate, positive correlation, and -0.18, a negligible, negative correlation, for the ML1M and ML100K dataset respectively (see Figure 4.8). Applying the Diversity Tradeoff model optimisation objectives to the ML1M dataset returns a negligible correlation coefficient of 0.01, demonstrating a complete lack of any correlation (see Figure 4.9). In contrast, the ML100K dataset shows moderate, negative correlation with a score of -0.61. The Compositional Fairness model shows the highest level of correlation for the BX dataset, with a score of 0.79. The ML1M dataset produces a correlation coefficient of -0.30 and the ML100K dataset returns a score of 0.71 (see Figure 4.10). The corresponding scatter plot can be found in the appendix (see Figure B.5). Similarly to the previous paragraph, we cannot deduce a universally applicable relationship between the two tested evaluation measures. The nature of their relationship is highly dependent on the underlying dataset and applied recommendation algorithm.

CV_{RMSE}* vs *CV_{VU} After a first explanation as to why the two user unfairness evaluation measures do not necessarily need to correlate with each other (see Section 4.5.1) we now want to further evaluate their relationship in dependence of the used recommender models and underlying datasets. Evaluating both recommender models' optimisation objectives on the ML1M dataset shows a moderate, positive correlation, with a score of 0.62 between the CV_{RMSE} and CV_{VU} evalua-

tion measure. However, this observation does not translate to the ML100K dataset, which produces a negligible, negative correlation with a coefficient of -0.15 (see Figure 4.8). Both datasets produce similar levels of correlation when evaluating only the results of the Diversity Tradeoff model. With correlation coefficients of -0.25 and -0.26, for the ML1M and ML100K dataset, the Diversity Tradeoff model shows only negligible correlation between the two user fairness evaluation measures. Similarly, the Compositional Fairness model also shows negligible to low correlation, with a correlation coefficient of -0.33 for the ML1M dataset and a correlation coefficient of 0.29 for the ML100K dataset. Only the BX dataset produces a higher correlation with a high, positive correlation coefficient of 0.73. The scatter plot depicting the distribution of results for the individual optimisation objectives can be found in the appendix (see Figure B.6). Again, the distinctiveness of the relationship is dependent on the underlying dataset and applied recommendation model.

AD@20* vs *CV_{RMSE} Calculating the correlation coefficients between aggregated diversity and user fairness measured by CV_{RMSE} on the optimisation results of both recommender models, returns a low, negative correlation score of -0.40 for the ML1M dataset and a negligible, positive correlation score of 0.15 for the ML100K dataset. Evaluating only the Diversity Tradeoff model training objectives shows a negligible correlation of -0.06 and a moderate, positive correlation of 0.69 for the ML1M and ML100K dataset respectively. The correlation results for the Compositional Fairness model are similar in difference of correlation distinctiveness between the two datasets. While the ML1M dataset shows a low, negative correlation score of -0.48, the ML100K dataset produces a high, negative correlation score of -0.75. Similarly to the ML100K dataset, applying the Compositional Fairness model to the BX dataset leads to a high, negative correlation coefficient of -0.84. These differences in correlation between the datasets can be further analysed by looking at the scatter plot shown in Figure B.4. For the Diversity Tradeoff model, the ML1M dataset produces more variant results than the ML100K dataset when optimising for individual diversity. Similarly, the ML1M dataset also shows more variance in results for the user fairness optimisation objective than the other datasets. The ML100K dataset demonstrates contrasting moderate to high correlations between aggregated diversity and user unfairness measured by CV_{RMSE} for the two applied recommender algorithms. The BX dataset shows a similar level correlation for the Compositional Fairness model.

RMSE* vs *CV_{RMSE} The relationship between *RMSE* and CV_{RMSE} shows a negligible to low negative correlation for both the ML1M and ML100K dataset when considering all training objectives. The Diversity Tradeoff model applied to the ML1M dataset returns a moderate, negative correlation with a score of -0.62 and the ML100K dataset produces a negligible correlation. Very contrasting behaviour is shown by the results gained from the Compositional Fairness model. Applied to the ML1M dataset, the results show negligible correlation for the *RMSE* and CV_{RMSE} correlation coefficient, while applied to the ML100K dataset, the model produces a moderate negative correlation with a correlation coefficient of -0.66 (see Figure 4.10). Similarly to the ML100K dataset, applying the Compositional Fairness model to the BX dataset leads to a high, negative correlation

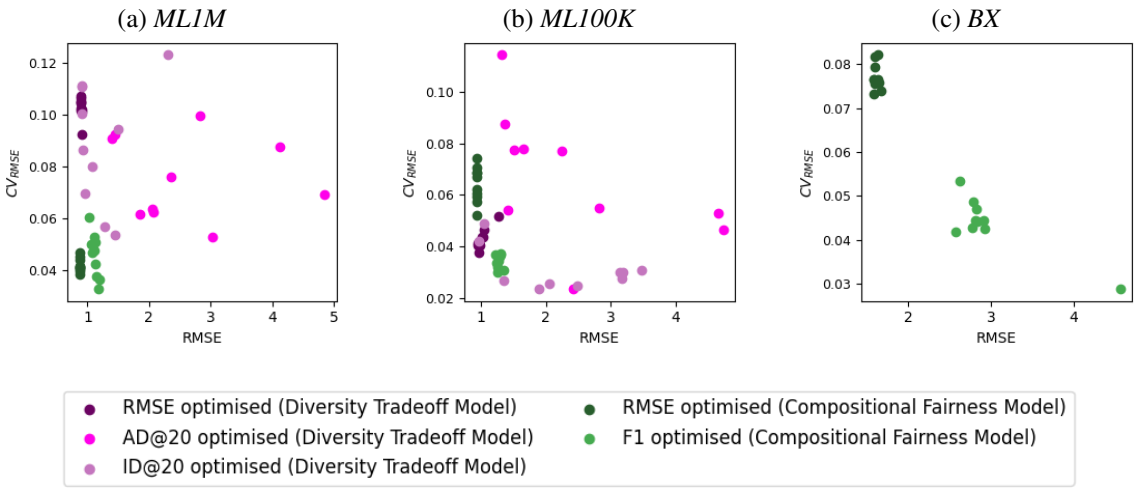


Figure 4.17.: ***Scatter plots for prediction performance and user unfairness (CV_{RMSE} over all optimisation objectives.***

with a score of -0.81. For the ML100K and BX dataset, a decrease in prediction performance goes along with an increase in user fairness measured by CV_{RMSE} and vice versa. This difference in correlation between the two evaluation measures depending on the underlying dataset can also be seen in the scatter plot shown in Figure 4.17. In the case of the ML1M dataset, the overall twenty data points belonging to the optimisation objectives of the Compositional Fairness model show no approximately uniform development. Evaluating the data points produced by the ML100K and BX datasets, we can see a more monotonic relationship for both optimisation objectives. Higher scores of user unfairness measured by CV_{RMSE} tend to relate to lower scores of prediction performance measured by *RMSE*. A similar pattern w.r.t. the ML1M Dataset can also be observed for the Diversity Tradeoff model optimisation objectives. Most data points from model instantiations optimised on prediction performance (measured by *RMSE*) and individual diversity (measured by *ID@20*) show a monotonic, ranked relationship, where a decrease in CV_{RMSE} goes along with an increase in *RMSE*. The Diversity Tradeoff model optimised on aggregated diversity (measured by *AD@20*) does not adhere to this pattern for the ML1M dataset.

RMSE* vs *ARP@20 The correlation matrices calculated over both models' results (see Figure 4.8) show only negligible correlation between prediction performance measured by *RMSE* and item popularity bias measured by *ARP*@20. Similarly, having a closer look at just the Diversity Tradeoff model, we also see a low, negative correlation with a score of -0.38 for the ML1M dataset and a negligible correlation for the ML100K dataset. Only the results obtained by the Compositional Fairness model show a higher correlation. The model demonstrates a moderate to high, negative correlation between the *RMSE* and *ARP@20* evaluation measure (see Figure 4.10). The correlation values for the ML1M and ML100K datasets are very similar to those between the *RMSE* and *ID@20* evaluation measures, which can be explained by the very high positive correlation between *ID@20* and *ARP@20*. The correlation matrix (see Figure 4.10) shows a moderate,

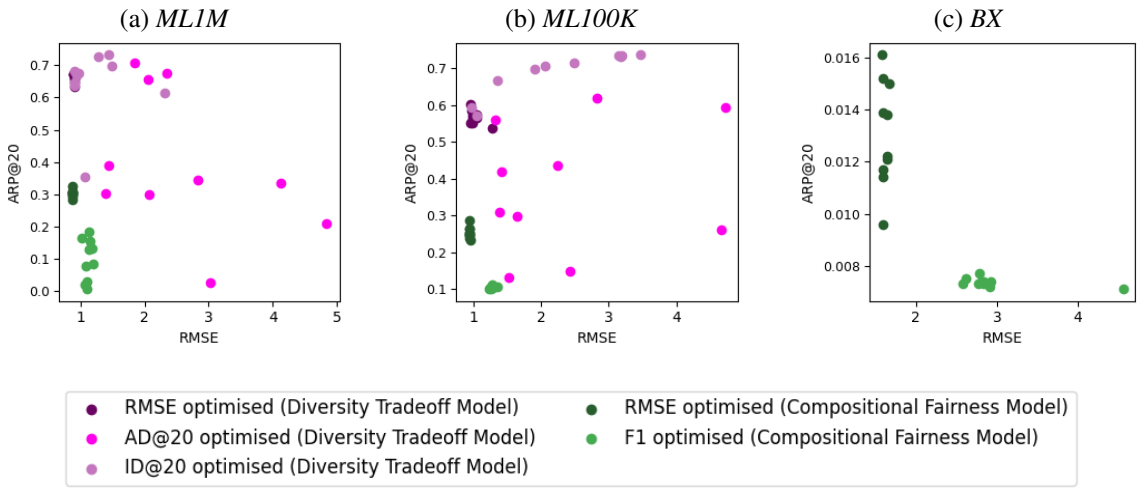


Figure 4.18.: ***Scatter plots for prediction performance and popularity bias over all optimisation objectives.***

negative correlation between *RMSE* and *ARP@20* for the ML1M dataset with a score of -0.69 and a high, negative correlation for the ML100K dataset with a score of -0.74. The highest correlation using the Compositional Fairness model is achieved by using the BX dataset, which produces a high, negative correlation coefficient of -0.84. Figure 4.18 shows the distribution of the data points for each dataset. While the data points obtained from the Compositional Fairness model show a predominantly ranked relationship between the *RMSE* and *ARP@20* evaluation measure, the Diversity Tradeoff model produces results of too much variance to identify any reliable type of relationship.

RMSE* vs *ID@20 The correlation between prediction performance and individual diversity, measured by *RMSE* and *ID@20*, can be neglected for the cross-model as well as Diversity Tradeoff model correlation calculation. The Compositional Fairness model shows a moderate to high negative correlation for the two evaluation measures for both the ML1M and ML100K dataset. The ML1M dataset produces a moderate, negative correlation of -0.68 and the ML100K dataset a high, negative correlation of -0.73 (see Figure 4.10). Therefore, when using the Compositional Fairness model, prediction performance and individual diversity are contradictory to each other. An increase in prediction performance measured by *RMSE* goes along with a decrease in individual diversity measured by *ID@20* and vice versa. The scatter plot showing the distribution of data points can be found in the appendix (see Figure B.1).

RMSE* vs *AD@20 Similarly to the relationship between *RMSE* and *ID@20*, evaluating the relationship between prediction performance and aggregated diversity using the *RMSE* and *AD@20* scores shows only negligible correlation for the overall correlation calculation (see Figure 4.8) and the Diversity Tradeoff model calculation (see Figure 4.9). This result holds true for both the ML1M and the ML100K dataset. Examining the results of the compositional Fairness model,

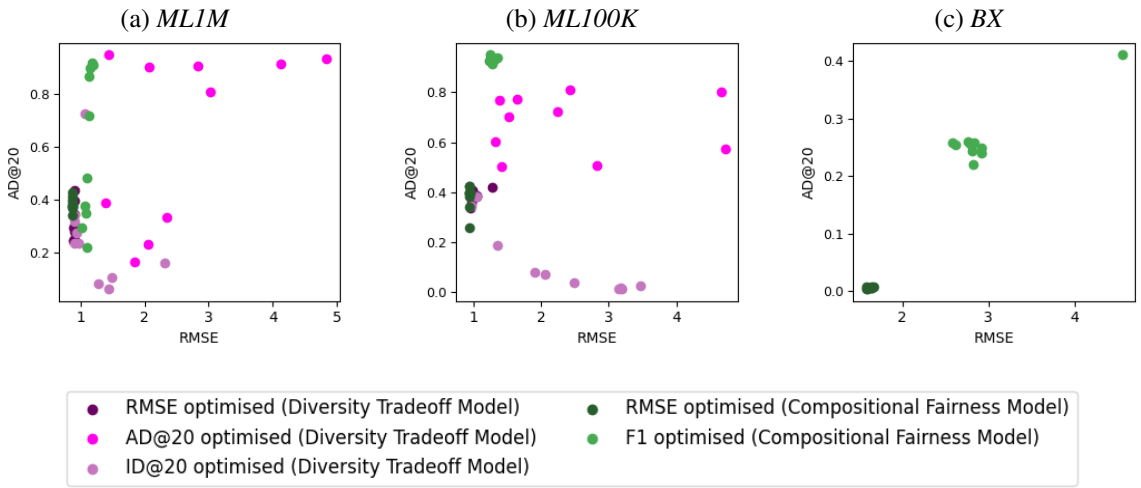


Figure 4.19.: ***Scatter plots for prediction performance and aggregated diversity over all optimisation objectives.***

shows a different relationship between the two evaluation measures. While the correlation is still low using the ML1M dataset, the ML100K dataset produces a high, positive correlation with a score of 0.75 between *RMSE* and *AD@20*. Using the BX dataset also produces a high, positive correlation of 0.81 between the two evaluation measures (see Figure 4.10). The scatter plot in Figure 4.19 shows the distribution of data points for the different datasets. Optimising the Compositional Fairness model on its two training objectives using the ML1M dataset, leads to an overlapping distribution of *AD@*20 scores between the two objectives, negatively impacting the correlation.

4.3.1. Summary and interpretation of results

The level of correlation between recommendation objectives that we have observed was often strongly influenced by the choice of recommendation model and by the underlying dataset. Despite this variance, the individual models and datasets showed similarities in correlations for certain recommendation objective combinations. We could also observe that the influences of the different training objectives detected in RQ1 do not necessarily guarantee a correlation between the respective recommendation objectives. The by far highest and only correlation demonstrated by each dataset and recommendation model, was shown between the individual diversity and popularity bias in recommendations. The two objectives share a strong, positive relationship, as both develop in identical directions during changes. As we have already examined earlier, a positive relationship between individual diversity and popularity is not expected due to popular items tending to show higher levels of similarity simply due to their high number of user interactions. We were able to prove this correlation, regardless of the underlying dataset and used recommendation model, similarly to the results obtained in RQ1. We also proved the predominant existence of a moderate to high, negative correlation between aggregated and individual diversity for both rec-

ommender models. Due to the very strong, positive correlation between individual diversity and popularity bias, aggregated diversity and popularity bias show a very similar level of correlation. An increase in aggregated diversity thus tends to go along with a decrease in individual diversity and popularity bias, and vice versa. These observed correlations coincide with our findings from RQ1, where we proved a positive influence of increasing individual diversity on popularity bias and its negative influence on aggregated diversity. Similarly, the negative influence of increasing aggregated diversity on individual diversity and popularity bias also reflected in the calculated correlations. The ML1M dataset used in combination with the Compositional Fairness model is the only combination not producing strong, negative correlations between aggregated diversity and individual diversity, as well as between aggregated diversity and popularity bias. This can be explained by the high level of variance w.r.t. the $AD@20$ score produced by the Compositional Fairness model optimised on user fairness using the ML1M dataset (see Figure 4.3b) while the other datasets show only low levels of variance for the model. Focussing on exclusively the cross-model correlation scores, we saw a moderate to high, positive correlation between individual diversity and user unfairness measured by CV_{VU} as well as between popularity bias and CV_{VU}. The higher the user unfairness, the higher the individual diversity and popularity bias. This finding is consistent with the insights we have gained from examining the influence of the individual diversity-focused training objective on the level of user unfairness measures by CV_{VU}.

The Diversity Tradeoff model showed no additional dataset-independent correlations between recommendation objectives, but the model exhibited strong differences in correlation levels between the ML1M and ML100K dataset. As an example, while the ML100K dataset showed a strong, positive correlation between aggregated diversity and user unfairness measured by CV_{RMSE}, the ML1M dataset did not show any type of correlation between the two recommendation objectives. This difference might be due to the ML1M dataset producing more variant CV_{RMSE} scores.

The Compositional Fairness model demonstrated a few additional model-specific correlations between different recommendation objectives. We proved the existence of a moderate to high, negative correlation between prediction performance and individual diversity as well as between prediction performance and popularity bias. A decrease in prediction performance (growing $RMSE$) tended to go along with an increase in individual diversity and popularity bias, and vice versa. These correlations completely disappeared in the cross-model correlation results, and could also not be detected in the Diversity Tradeoff model. While investigating RQ1 we showed that optimising on increased individual diversity led to a decrease in prediction performance overall tested dataset, due to a high level of variance produced by the Diversity Tradeoff model, we were not able to prove an overall correlation between the two objectives. Additionally, the Compositional Fairness model also demonstrated a moderate to high, negative correlation between prediction performance and user unfairness measured by CV_{VU} while the Diversity Tradeoff model and the cross-model results only showed low to moderate, positive correlation. Similarly to the Diversity Tradeoff model, the Compositional Fairness model also suffered from varying correlation levels depending on the used dataset. The BX dataset returned almost exclusively high-value correla-

tions for all combinations of recommendation objectives, likely due to the low levels of variance produced by the dataset w.r.t. the individual evaluation measures (shown by the boxplot diagrams in Section 4 and by the scatter plots shown in Section 4.2.1). The correlation matrix of the ML100K dataset obtained from the Compositional Fairness model is very similar to that of the BX dataset, albeit less pronounced. The ML1M dataset showed a few mutually exclusive correlations in comparison to the ML100K and BX dataset. While the ML1M dataset showed only negligible correlations between aggregated diversity and individual diversity as well as popularity bias, the other two datasets produced strong, negative correlations between these recommendation objectives. Similarly, the ML1M dataset used for the Compositional Fairness model produced negligible and low, negative correlation coefficients for all correlations with the CV_{RMSE} evaluation measure while the other datasets produce multiple contrasting negative correlations for the CV_{RMSE} measure. This further supports our assumption that differences in datasets have strong influence on the variance of evaluation results and inevitably also on the correlations. While the ML1M and BX dataset produced high, negative correlation coefficients for the relationship between individual diversity and user unfairness measures by CV_{VU}, the ML100K and BX dataset showed a high, positive correlation between aggregated diversity and individual diversity. These dataset-dependent variances show again, that there seems to be no universally applicable truth with regards to relationships between recommendation objectives but that the results highly depend on the underlying data. Using scatter plots as means to visualise the distribution of results over the datasets, we could also show that certain optimisation objectives tended to show no coherent spread of values while others presented correlating behaviour for most combinations of objectives. One example of the latter is the Diversity Tradeoff model optimised on individual diversity, which shows higher levels of correlations for a high number of correlation analyses. The Diversity Tradeoff model optimised for aggregated diversity, on the other hand, showed a high amount of incoherent spread for many combinations of recommendation objectives during correlation analysis, proving again, that optimising on non-prediction performance objectives leads to less stability of the model. The correlation between recommendation objectives is also characterised by a dependence on the used recommender model. While we were able to point out a few universally applicable correlations, we have also shown that these correlations are often heavily influenced by the applied recommendation algorithm and by the underlying dataset. Stronger variances over multiple optimisation objectives for an evaluation measure are a reason for lower correlation values, as this leads to the values not allowing for a ranked or even linear relationship. With reference to the dataset, possible reasons for these differences could be the varying ratios between users and items or the differences in density of the datasets.

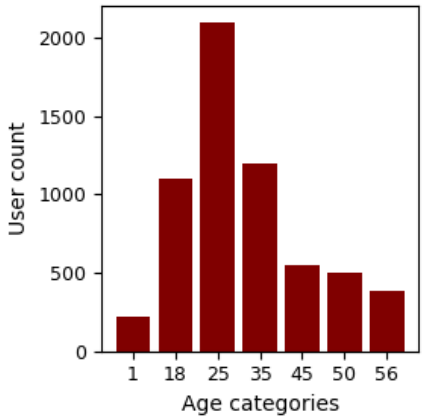


Figure 4.20.: ***Age distribution over the ML1M training data***

age group	1	18	25	35	45	50	56	∅	weight ∅	σ
∅ rating	3.5548	3.4952	3.5476	3.6143	3.6573	3.6905	3.7617	3.6173	3.5866	0.0857
∅ pred γ=0	3.6021 (+0.0473)	3.5040 (+0.0088)	3.5478 (+0.0002)	3.6179 (+0.0036)	3.6235 (-0.0338)	3.7038 (0.0133)	3.7525 (-0.0092)	3.6216	3.5882	0.0788
∅ pred γ=1000	3.4660 (-0.0888)	3.4826 (-0.0126)	3.5490 (+0.0014)	3.5916 (-0.0227)	3.5510 (-0.1063)	3.6272 (-0.0633)	3.6404 (-0.1213)	3.5582	3.5545	0.0620

Table 4.4.: ***Influence of the fairness objective on rating predictions over age groups.*** *With increasing fairness penalty, the predicted ratings show an alignment in height for the different user categories when using the ML1M dataset.*

4.5. Case Studies

4.5.1. Case study A - Understanding the definition of fairness for the Compositional Fairness model

As already introduced in Section 2.3.4, there is no universal definition of user fairness when working with recommender systems. One definition characterises a recommender system as user fair if it offers equal recommendation performance to users irrespective of their affiliation with different user subgroups. The CV_{RMSE} score evaluates how much variance in prediction performance is produced by a recommender system across different user groups (see Section 2.3.4). Another approach proposes the definition of user fairness as a recommender system making no distinction between user groups and predicting similar ratings across all groups. This can be measured by calculating the CV_{VU} score as proposed in Section 2.3.4. To understand the definition of fairness that the Compositional Fairness model uses, table 4.4 shows the influence that increasing the fairness objective has on rating predictions across the different user groups. The rating values for the $\varnothing rating$ have been calculated based on the training data used for training the Compositional Fairness model. The rating predictions for both the model instantiation using no fairness objective ($\gamma = 0$) and the model instantiation using the maximum fairness objective ($\gamma = 1000$) have been obtained using one Compositional Fairness model instantiation, optimised for prediction perfor-

mance for the $\gamma = 0$ predictions and only varying the γ parameter, controlling the fairness penalty, during training the second instantiation. Doing some descriptive evaluation of the data, we can see an overall positive correlation between the age group value and the average ratings calculated on the training data. On average, older users have given higher movie ratings than younger users. To get a numerical representation of this difference in rating behaviour, we calculate the standard deviation over all averaged rating values. Comparing the standard deviation of the training data with prediction results using different levels of fairness penalty γ allows us to analyse whether an increase in fairness penalty goes along with a decrease in prediction variance across age groups. Calculating the average rating predictions using no fairness penalty ($\gamma = 0$) shows that for many age categories, the model can closely match its prediction range to the original ground truth values (values in brackets show the difference to the ground truth). It is noticeable that the predicted rating values for the lowest rating (category 18) as well as for the highest rating age category (category 56) decrease in distance to each other. The standard deviation over all age categories predicted rating averages decreases slightly to a value of 0.0788 in comparison to the ground truth rating results ($\sigma = 0.0857$). Noticeable is that the three age categories with the most users in the training data (categories 18, 25, 35) (cf. figure 4.20) show the least change in average rating values. Increasing the fairness penalty to the maximum ($\gamma = 1000$) leads to a noticeable decrease in the standard deviation over rating values of all age categories ($\sigma = 0.0620$). This behaviour can also be seen in the distribution of VU scores of the individual user groups (see Figure 4.21. With increasing fairness penalty γ, the spread of the VU scores of the different age groups decreases. This indicates a decrease in user unfairness according to the CV_{VU} score. The three user age categories most frequently represented in the training data (categories 18, 25, 35) still show the least deviation in average rating predictions compared to the ground truth. We can also see that the average ($\varnothing$) and the weighted average over all ratings ($weight\varnothing$) are now very close to the ground truth rating score for age category 25, which is the category with by far the most users. While the standard deviation over average ratings decreases, the difference of average rating predictions to the averaged ground truth ratings increases with varying degrees of the age categories. This development might imply an increase in user unfairness according to the CV_{RMSE} score. This claim can be supported by comparing the $RMSE$ scores of each individual age category (see Figure 4.21). Increasing the fairness penalty γ leads to a decrease in user unfairness measured by CV_{VU} and an increase in user unfairness measured by CV_{RMSE}. The results for evaluating the influence of the fairness penalty using the ML100K dataset and the BX dataset can be seen in Figures B.7, B.8. The three datasets show differences in the development of the $RMSE$ and VU scores of the different age categories. Increasing the fairness penalty on the Ml100K dataset and the BX dataset shows a decrease in user unfairness measured by CV_{VU} as well as a decrease in user unfairness measured by CV_{RMSE}.

Due to the fact, that this analysis only uses results produced by using only one training seed per dataset and the variance of outcomes, the reliability of these results is not given. According to the results presented in this case study, we tend to interpret the fairness objective implemented by

(a) *RMSE scores of the individual age categories* (b) *VU scores of the individual age categories*

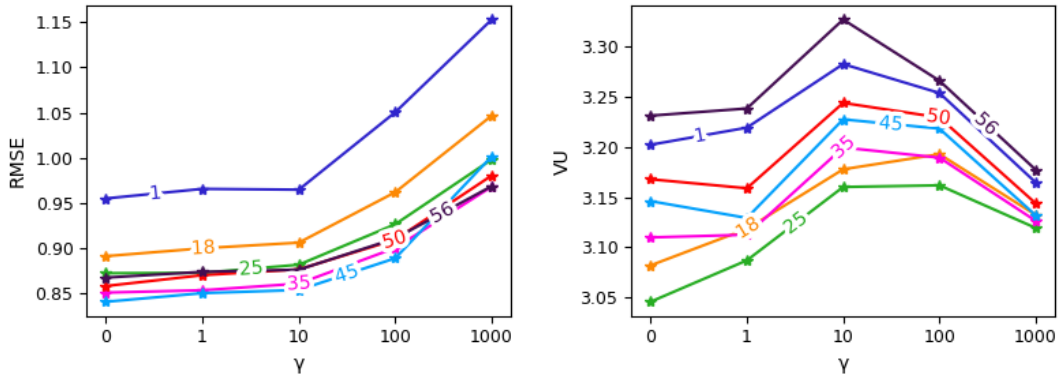


Figure 4.21.: ***The influence of varying the fairness objective on the Compositional Fairness model on the** RMSE **and** VU **scores of the individual age categories.*** *The results have been produced using the ML1M dataset and are based on one instantiation optimised for prediction performance, only changing the γ setting for instantiations. With increasing fairness penalty the RMSE scores across age categories increase in variance while the VU scores decrease in variance.*

[5] for the Compositional Fairness model as being similar to the CV_{VU} approach we have adapted from Yao and Huang [25]. With increasing fairness penalty, the standard deviation over average ratings of user categories is decreasing, indicating an alignment of predicted ratings over all user categories.

4.5.2. Case study B - Influence of the fairness penalty on the performance of the Compositional Fairness model

This case study evaluates the influence of the fairness penalty for the Compositional Fairness model by using one instantiation of the Compositional Fairness model optimised for prediction performance and one optimised for user fairness per dataset. Using these instantiations as a starting point, we vary the value of the fairness penalty γ while keeping all other parameter settings, including the training seed, identical. The higher the value of γ, the stronger the fairness penalty applied during training. These results only offer a first impression of how the fairness penalty might incrementally influence the other evaluation measures. Figure 4.22a shows the influence that varying the fairness penalty of the Compositional Fairness model has on the prediction performance measured by the *RMSE* score. We can see that all graphs, regardless of underlying dataset and optimisation objective, show an overall growing trend w.r.t. the *RMSE* score. Increasing the fairness penalty thus leads to a decrease in prediction performance. Figure 4.22b shows the influence of the γ fairness penalty on the level of popularity bias in the recommendation results. Due to the overall very low level of popularity bias in the BX dataset, the γ variation has the comparably

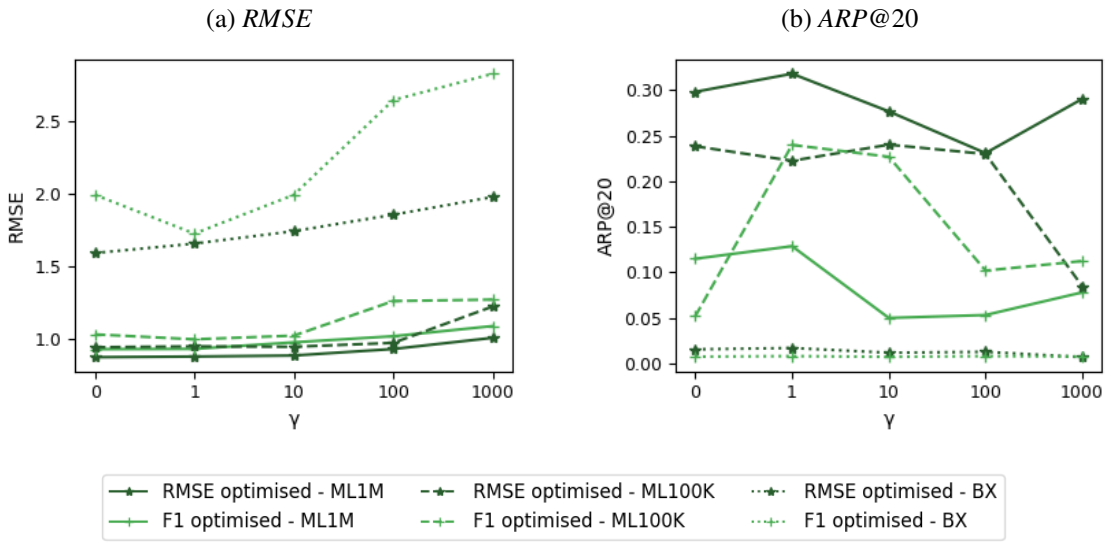


Figure 4.22.: ***The influence of varying the fairness objective on the Compositional Fairness model on prediction performance and popularity bias.***

the least influence on the BX dataset's results. Interestingly, the prediction performance-optimised plots for the ML1M and ML100K dataset show contrasting behaviour with increasing fairness penalty. The results obtained from optimising on user fairness on the ML1M and ML100K dataset show fairly similar behaviour for varying values of γ, albeit with different levels of strength. They reach their highest level of popularity bias for a fairness penalty value of $\gamma = 1$.

Looking at the development of the level of aggregated diversity in recommendation results (see Figure 4.23a), we can see that, with one exception, the optimisation objectives produce higher aggregated diversity when trained with a fairness penalty of $\gamma = 1000$ than when trained using a fairness penalty of $\gamma = 0$. The strongest increase in aggregated diversity can be observed using the ML100K dataset. Evaluating the individual diversity (see Figure 4.23b), we can only compare the results of the ML1M and ML100K dataset. The models optimised on user fairness show fairly similar development over varying levels of fairness penalty, while the models optimised on prediction performance show contrasting behaviour for the two datasets, similarly to the plot behaviour shown in the popularity bias evaluation.

Comparing the development of the $F1$ and CV_{RMSE} scores shown in figures B.9a, 4.24a, we can see a few similarities between the behaviour of the two user unfairness evaluation measures. Except for the prediction performance-optimised instantiations trained on the ML100K and BX dataset, the models reach their lowest CV_{RMSE} score at the same fairness penalty level, that they reach their lowest $F1$ score at. The prediction performance and user fairness optimisation objectives trained on the ML1M dataset show similar behaviour over the different values of γ, while the prediction performance optimisation objectives for the ML100K and BX dataset show similar development, same as the user fairness-focussed optimisation objectives for the two datasets. The plot shows an overall downward-oriented trend for the CV_{RMSE} score with increasing fairness

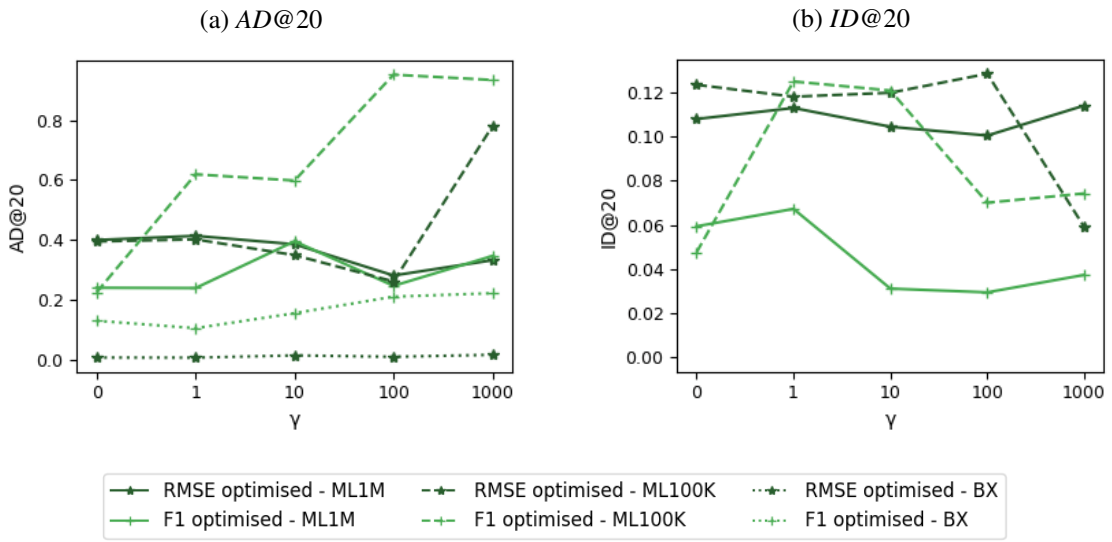


Figure 4.23.: ***The influence of varying the fairness objective on the Compositional Fairness model on aggregated and individual diversity.***

penalty. Figure 4.24b shows how the fairness penalty influences the level of user unfairness measured by CV_{VU}. The graph shows an overall decreasing trend of user unfairness with increasing fairness penalty, even if not every model instantiation reaches its lowest unfairness with the maximum fairness penalty of $\gamma = 1000$. The CV_{RMSE} and CV_{VU} plots show no uniform behaviour, with some instantiations showing contrasting and some showing comparable behaviour.

4.5.3. Case study C - Influence of the Furthest Neighbour graph on the performance of the Diversity Tradeoff model

In this case study, we have a brief look at how the FN graph influences the evaluation results obtained from the Diversity Tradeoff model. We choose one instantiation for every optimisation objective per dataset and run the recommendation model using identical parameter settings, only varying the α parameter, which influences the impact of the FN graph. The higher the α value, the stronger the influence of the FN graph during training in comparison to the NN graph. Important is that not all evaluated models use the same graphs for the FN calculation. While they all use an item-based NN graph, some use an item-based and some a user-based FN graph. Starting with the prediction performance evaluation, we can see that the performance optimised model is the only model showing no changes influenced by a varying level of FN graph influence during training, regardless of the underlying dataset. Using the ML1M dataset, the optimisation objectives for aggregated and individual diversity show decreasing prediction performance with increasing influence of the FN graph. Optimising for individual diversity on the ML100K dataset shows a slight linear decrease regarding the performance, and optimising for aggregated diversity produces

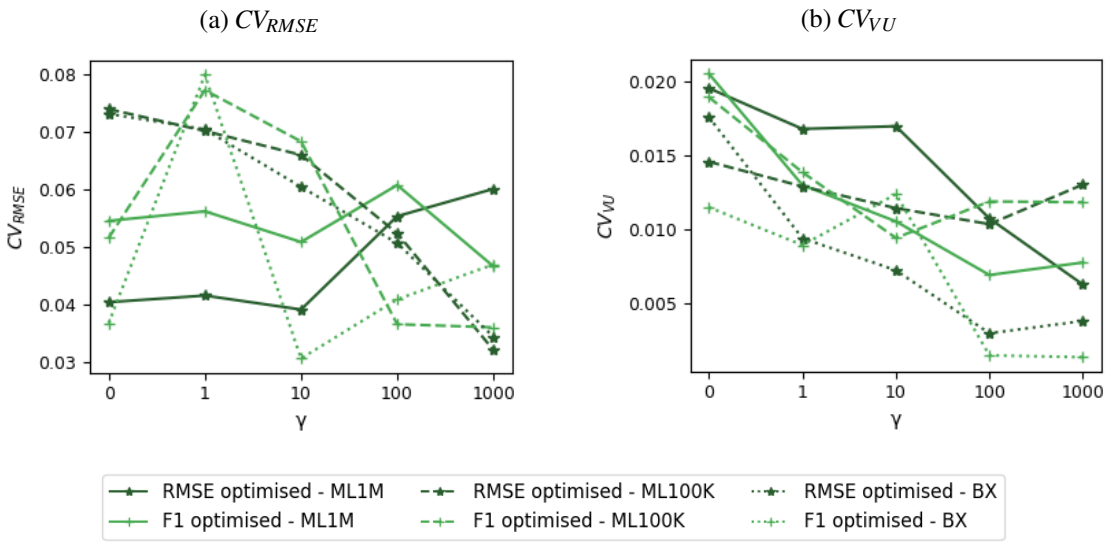


Figure 4.24.: ***The influence of varying the fairness objective on the Compositional Fairness model on user fairness measured by CV_{RMSE} and CV_{VU}.***

the best results for moderate to high values of α. Evaluating the influence of the FN graph on the level of popularity bias (see Figure 4.25b), we can see that the models optimised on prediction performance and individual diversity show only small variance w.r.t. the level of popularity bias with increasing α. Optimising on aggregated diversity, on the other hand, shows an overall increase in popularity bias with increasing influence of the FN graph during training. The model optimised on the ML1M dataset shows a slight decrease when reaching the highest values of α.

Next, we are evaluating the influence of the FN graph involvement during training on the level of aggregated diversity in recommendation results. The first observation we can make on the plotted data in figure 4.26a is that the results of the different optimisation objectives only show small changes regarding the distinctiveness of the aggregated diversity. The results obtained from the ML100K dataset show overall less variance than the results obtained from the ML1M dataset. The graph depicting the levels of individual diversity 4.26b shows similar lack of variance in results for the models optimised for prediction performance and individual diversity. The development of the individual diversity in recommendation results depending on the influence of the FN graph for the models optimised on aggregated diversity differs between the two used datasets. While the ML1M dataset shows a slight increase in individual diversity before decreasing with higher α values, the ML100K dataset produces a slight decrease in individual diversity for low α values before increasing considerably with increasing influence of the FN graph.

When it comes to evaluating the level of user unfairness in recommendation results (see Figure B.10), the first observation we can make is the seeming absence of any universal structure regrading the development of the CV_{RMSE} score. Optimising on prediction performance, regardless of the underlying dataset, shows the least changes in user fairness according to the CV_{RMSE} and CV_{VU} scores when increasing the influence of the FN graph during training. The CV_{RMSE} scores pro-

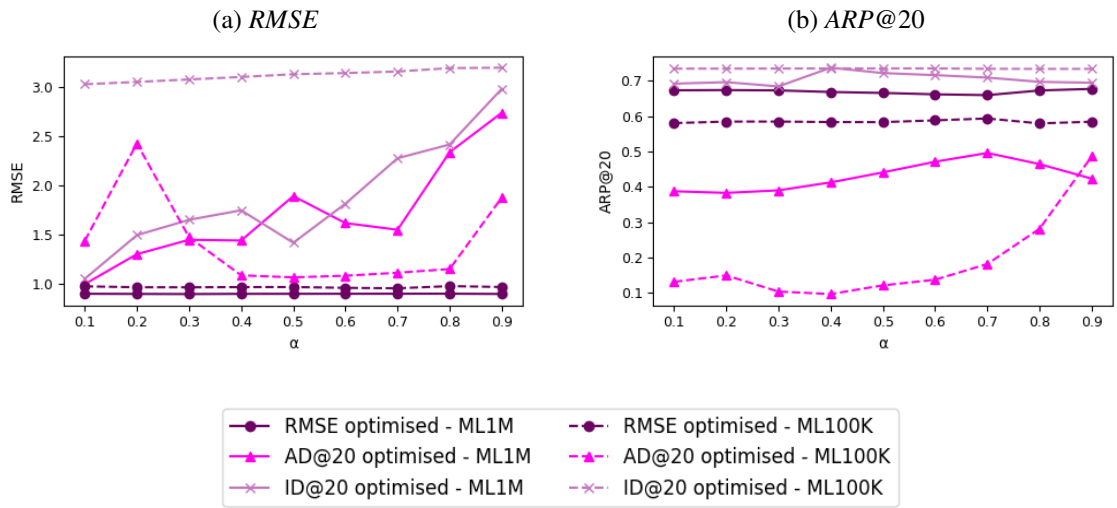


Figure 4.25.: ***The influence of varying the impact of the FN graph in the Diversity Tradeoff model on prediction performance and popularity bias.***

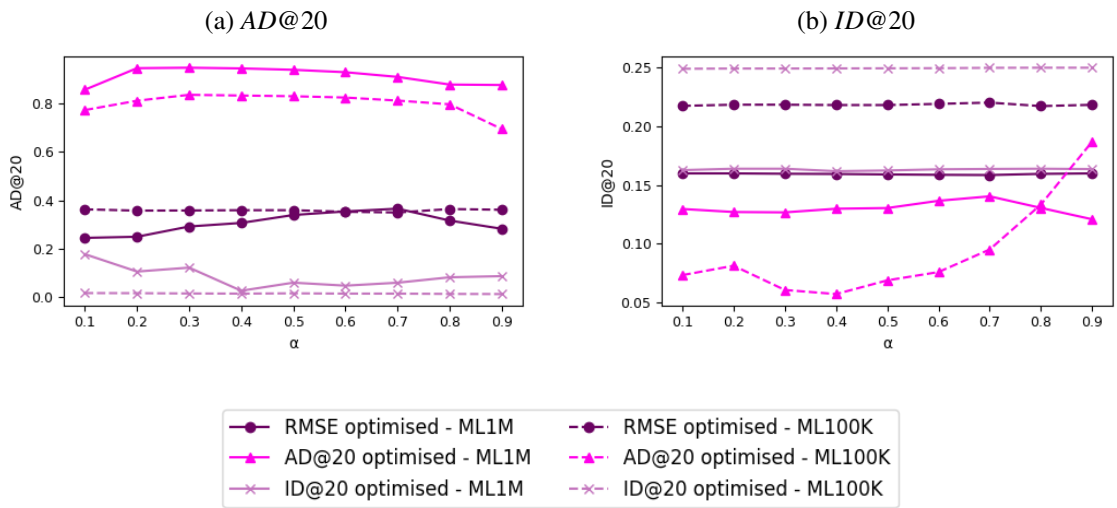


Figure 4.26.: ***The influence of varying the impact of the FN graph in the Diversity Tradeoff model on aggregated and individual diversity.***

duced by optimising on aggregated diversity show rather contrasting behaviour when comparing the two datasets. For $\alpha = 0.2$, the ML1M dataset reaches its highest level of unfairness, while the ML100K dataset reaches its lowest level of unfairness. Using the using ML100K dataset shows no variance for the CV_{RMSE} score when modifying the α parameter, while the ML1M dataset shows a higher level of variance which shows a high similarity to the development of the CV_{VU} score obtained from the same dataset and optimisation objective. The ML100K and ML1M dataset optimised on individual diversity show a mostly increasing level of user unfairness measured by CV_{VU} similarly to the ML1M dataset optimised on aggregated diversity.

5.

Related work

To put the research conducted in this thesis into context with the current state of research in the area of interdependency of recommendation objectives and prediction instability in recommender systems, the following chapter introduces a few related research papers. To the best of our knowledge, no prior research has focused explicitly on graph-based recommender systems in this context.

Beyond-accuracy: Aggregated and individual diversity

Kaminskas and Bridge [13] analyse the relationship between recommendation accuracy, aggregated diversity, individual diversity as well as serendipity and novelty in recommendation results. In contrast to our work, the authors included the additional item-focused recommendation objectives serendipity and novelty into their research, while not evaluating user fairness and prediction instability of recommendations. Kaminskas and Bridge [13] measure prediction performance using the *recall* score, a metric evaluating the relevance of ranked recommendations, the individual diversity using a rating-based distance metric, and aggregated diversity based on the catalogue coverage of the recommendation results. The authors use four, at that time, state-of-the-art recommendation algorithms for their evaluation of relationships between the different recommendation objectives. They divide their research in two parts. The first part focuses on the relationship between individual recommendation objectives when comparing the results of different recommendation algorithms optimised on prediction performance. Using two datasets for testing, the authors prove the existence of positively correlated behaviour between prediction performance, aggregated diversity and individual diversity. For most tested recommenders and datasets, an increase in prediction performance goes along with an increase in individual as well as aggregated diversity. The second part of their research aims at evaluating relationships between recommendation objectives through re-ranking the recommendation results for individual diversity, serendipity, or novelty. The authors individually apply a re-ranking approach for most tested recommendation objectives to the recommendation results achieved by each recommendation algorithm. The results demonstrate, that applying a re-ranking approach for individual diversity correlates with an increase in

novelty and an increase of aggregated diversity in recommendation results. Overall, the authors deduce that re-ranking the recommendation results using a non-prediction performance-oriented objective leads to a decrease in prediction performance.

An analysis of recommendation biases

Jannach et al. [12] evaluate the tradeoff between aggregated diversity and popularity bias during the training cycle of recommendation algorithms. While the authors do not compare the effect that different optimisation objectives have on the other involved recommendation objectives, they provide a first overview of possible correlations between objectives based on their evolution over time. Jannach et al. [12] measure the prediction performance using the *RMSE* score, the popularity bias using the *Gini* score, and aggregated diversity based on the catalogue coverage in the recommendation results. The individual diversity is calculated based on the intra-list similarity of users' top-k recommendation results. The authors use a *matrix factorisation recommendation algorithm* for evaluating the interdependencies between the recommendation objectives. While the prediction performance of the tested recommendation algorithm stays comparatively constant over all training steps, there is a high amount of change with regard to the levels of popularity bias and aggregated diversity during the training cycle. The model training starts off with a very high popularity bias, which slowly decreases over the training course. In stark contrast to this stands the development of the aggregated diversity, which starts at its lowest value and shows increasing behaviour. The individual diversity starts at its overall lowest value and demonstrates a steep increase during the first few training steps, before moderately decreasing and increasing again. Jannach et al. [12] use on exemplary recommendation algorithm to show that aggregated and individual diversity are not mutually exclusive, even if they show contrasting behaviour at times. Popularity bias and aggregated diversity, on the other hand, show exclusively contrasting behaviour.

The unfairness of popularity bias

Yalcin and Bilge [24] analyse the relationship of popularity bias and recommendation performance, while also considering the differences in the relationship between these two recommendation objectives for different user groups. Users are split into three different groups based on how many ratings they have submitted (profile size) and the authors evaluate the differences in evaluation results between these groups. The recommendation performance is measured by the *normalised Discounted Cumulative Gain nDCG*, which measures the accuracy of users' recommendation lists based on the ground truth rating values and their predicted rank in the recommendations. A higher value indicates a better recommendation performance. As a measure of popularity bias, the authors evaluate the distribution of short-head and long-tail items in the recommendation lists of users. The more unpopular long-tail items have been recommended, the smaller the popularity bias. Yalcin and Bilge [24] test several recommendation algorithms on four

different datasets. For two of the four datasets, the authors prove the existence of a negative correlation between prediction performance and the level of popularity bias in the recommendation results. Using the other two datasets, the level of correlation is highly dependent on the applied recommendation algorithm. Yalcin and Bilge [24] also show that for most datasets, users with a big profile size receive more accurate predictions than users with a smaller profile size. With regard to the level of popularity bias in user recommendations, the differences between the user groups are not as pronounced. Summarising, the authors prove that the correlation between prediction performance and popularity bias and the distinctiveness of each objective is highly dependent on the used dataset and also varies between the different recommendation algorithms.

Popularity bias and miscalibration

Abdollahpouri et al. [1] define a recommender system as user-unfair if the recommended items do not represent a user's tastes and interests according to the user's group affiliation. This evaluation measure bases user groups not on characteristics like gender or age, but on users' interest, for example in specific movie genres. User fairness is defined as the difference between ground truth interest representation and the interest representation of items recommended to the different user groups. The authors categorise users based on their interest in popular and unpopular movies. By doing this, they are able to evaluate how popularity bias impacts user groups with different levels of interest in popular and unpopular items and how it correlates with the miscalibration of recommended items. The authors define popularity bias as the difference between ground truth popularity distribution and the popularity distribution in recommendations. Using these definitions of user fairness and popularity bias, the authors examine the relationship between the two evaluation measures by evaluating the results of five different recommender algorithms. Abdollahpouri et al. [1] prove a distinct, positive correlation between popularity bias and user unfairness. The higher the level of popularity bias produced by a recommender model, the higher the user unfairness w.r.t. the miscalibration of user interests.

Bias amplification and preference representation

Mansoury et al. [17] evaluate the effect that the so-called Feedback Loop has on amplifying popularity bias and its effect on the level of aggregated diversity in recommendation results. They also analyse to what extent bias amplification affects the representation of user interest. The authors do not specify how exactly they measure popularity bias and aggregated diversity in the obtained recommendation results, and only mention that preference representations for users are based on the genres of movies they have rated. Mansoury et al. [17] evaluate the development of popularity bias and aggregated diversity over the training cycle using three different recommendation algorithms. The results for the popularity bias evaluation measure take into account the changes to item popularity levels created by recommendations during the previous training step. With increasing iterations, all three recommender algorithms show ascending values of popularity bias, indicating the existence of bias propagation for each model. Simultaneously, measuring the level

of aggregated diversity shows descending behaviour for all tested algorithms. Popularity bias and aggregated diversity show overall contrasting behaviour, indicating a negative correlation. As a next step, Mansoury et al. [17] analyse if the existence of bias amplification also shows any connection to how well the recommender algorithms are able to capture the interests of different user groups. To achieve this, they evaluate how much the representation of user interests deviates from the ground truth representation during the training cycle. The authors analyse the difference in interest deviation between female and male users. As most users in the tested dataset are male, the male users show less deviation from their group's representation during the whole training cycle than the female users, which indicates differences in interests between the two user groups and a tendency of the recommender algorithms to recommend more male-oriented items, thus producing unfair results. The deviation is steadily increasing for both user groups, but the deviation for female users shows a more pronounced deviation. The authors come to the conclusion that the tested recommender systems suffer from bias amplification, leading to increased popularity bias. The increase in popularity bias correlates with a decrease in aggregated diversity of recommendation results and an unequal (thus unfair) decline in preference representation across user genders.

6.

Discussion

In this chapter, we will first discuss the limitations of the research conducted in this thesis, before concluding our main findings and giving an outlook on possible future research topics.

6.1. Limitations

Computational resources The biggest limitation we encountered during our experiments was the high demand for computational resources by the Diversity Tradeoff model. Due to the use of Nearest Neighbour and Furthest Neighbour calculations during the training process, running the model turned out to be quite time-consuming, requiring a certain level of downsizing during parameter optimisation. While the authors provided case studies as starting points for the downsizing, these case studies were based on one specific dataset. Additionally, With increasing number of users and items in the underlying dataset, the demand for computational and time resources for NN and FN graph calculations became unfeasible. Computing resources also ended up being a bottleneck for the calculation of item similarities for the individual diversity in recommendations.

Variance of results and prediction instability The high levels of prediction instability and variance of results when optimising on non-prediction performance-oriented objectives showed low reliability of the results obtained in this thesis. Due to the limited number of resources, we were not able to test different training seeds for every combination of parameters during optimisation, which shows high probability that the optimised results used in this thesis do not truly represent the maximum possible optimisation. While the instability and variance of results is an insight in itself, it complicates the evaluation of interdependencies between different recommendation objectives. Especially, the variances measured for relationships between individual recommendation objectives when using different datasets showed low reliability of the results.

Universal applicability The extent of research done in this thesis does not allow for the results to be considered universally applicable. The results presented in this thesis merely offer

an insight on the nature of relationships between recommendation objectives when using the Diversity Tradeoff model and Compositional Fairness model on the mentioned datasets. The high variance of results between the tested recommender models and datasets makes it very probable that these variances also exist for other models and datasets. This assumption gets further support from the results presented in related research (see Chapter 5). While none of the related works have focussed on graph-based recommendation, many of them show similar weaknesses w.r.t. a lack of universal applicability.

6.2. Conclusions

The overall research goal of this thesis was to evaluate how different training objectives influence the interdependency of prediction performance, item biases and user fairness. This section serves the purpose of summarising the main findings of our experiments. We have used two different recommender models, one focussing on increasing diversity in recommendation results and one focussing on increasing user fairness, and we have tested these models on three different datasets to gather more comparable and insightful results. The main take-away of our experimental approach to the research questions is that while we can observe overall influences of different optimisation objectives on the different recommendation objectives independent of the underlying dataset, the nature of correlations between recommendation objectives is highly dependent on the used datasets and recommendation algorithm.

RQ1 - How are prediction performance, item biases and user fairness influenced by diversity- and fairness-oriented training objectives?

With the first research question, we examined how prediction performance, different item biases and user fairness are influenced by varying training objectives. Optimising the used recommender models on non-prediction performance-oriented objectives leads to a decrease in prediction performance and an increase in variance for most evaluation measures, making the results produced by the models less reliable. The aggregated diversity-oriented training objective and the user fairness-oriented training objectives seem to have similar influence on the item-related recommendation objectives, increasing the level of aggregated diversity in recommendation results, while decreasing the individual diversity and popularity bias. Optimising on individual diversity seems to lead to opposite influence on the item-based recommendation objectives. We can conclude that while there is a variance for one user unfairness recommendation objective depending on the used dataset, the results show overall equal results for the different datasets. Nevertheless, we cannot declare these results universally applicable due to only being tested on two different recommender algorithms and three datasets.

RQ2 - How do these training objectives influence the prediction instability of the respective models and the variance of recommendation objectives?

For the evaluation of the second research question, we took into consideration the levels of prediction instability, as well as the levels of variance shown by the results of the different evaluation measures. The overall conclusion we can draw from the results is that optimising on non-performance-oriented training objectives leads to an increase in prediction instability as well as an increase in variance of results for the individual evaluation measures. The consequence we can extract from these results is that optimising on non-performance-oriented objectives leads to a decrease in reproducibility and reliability of the underlying models.

RQ3 - Are there any correlations between prediction performance, item biases and user fairness?

To answer the third research question, we evaluated the correlations between individual recommendation objectives for each dataset to gain insights about their respective relationships. We not only analysed the correlation of values over both used recommender models, but also evaluated the correlations produced by each model individually. We were able to prove the unexpected existence of a strong, positive correlation between individual diversity and popularity bias over all recommender models and datasets. Similarly, we showed the overall existence of a moderate to high, negative correlation between the aggregated diversity and individual diversity as well as between the aggregated diversity and popularity bias in recommendation results. Most other correlations between recommendation objectives seem highly dependent on the underlying dataset and used recommendation model, allowing no inference about their universal applicability. Overall, we can conclude that the higher levels of variance of the non-prediction performance-oriented optimisations w.r.t. the evaluation scores seem to be a big reason for the lack of structure and correlation in some results.

Case studies

Case study A - Understanding the definition of fairness for the Compositional Fairness model Evaluating the definition of fairness introduced by the Compositional Fairness model, we have found out, that an increase in fairness penalty has a positive influence on the level of user unfairness measured by the CV_{VU} score. Based on similar developments over all datasets, we are inclined to conclude that the fairness definition by the Compositional Fairness model might be of similar nature to our proposed evaluation measure.

Case study B- How are the relationships between recommendation objectives influenced by incrementally increasing the fairness objective during performance-

and fairness-oriented training? We have evaluated the influence that the fairness penalty used by the Compositional Fairness model has on the distinctiveness of the recommendation objectives analysed in this thesis. By gradually increasing the fairness penalty, we discovered that this increase leads to an overall decrease in prediction performance and decrease in user unfairness measured by the CV_{VU} score. We could also deduce an overall increase of aggregated diversity for increasing fairness penalty. Results obtained for the evaluation of the other recommendation objectives show high variance between the different applied datasets, similar to results obtained for our other RQs. Due to the lack of universal definitions of user fairness, our proposed evaluation measure is not universally applicable in the whole context of enhancing user fairness in recommender systems.

Case study C - How are the relationships between recommendation objectives influenced by incrementally increasing the influence of dissimilarity-oriented graph during performance- and diversity-oriented training? To evaluate this case study, we made use of the Diversity Tradeoff model's ability to vary the influence of the FN graph during model training. We observed that, for a few of the evaluated recommendation objectives, the influence of the FN graph was highly dependent on the applied training objective and underlying dataset. The training objectives not focused on prediction performance seem to be more influenceable by the leverage of the FN graph. For some recommendation objectives, the influence of the FN graph is also dependent on the underlying dataset. Interestingly, we could not deduce a universal impact of the FN graph on the levels of diversity in recommendation results. Increasing the influence of the FN graph during optimising for maximum individual diversity, even increased the popularity bias and decreased the aggregated diversity of recommendation results.

6.3. Future research

To further evaluate and understand the interdependency of biases, fairness and prediction instability, the following section provides a brief outlook on possible future research topics connected to the research done in this thesis.

Evaluation of universal validity of interdependencies Due to the limited generalisation ability of the conducted experiments, further experiments with graph-based Recommender Systems and additional datasets might give more insights with regards to the factors influencing the relationships between different recommendation objectives.

Influence of GNNs on the relationships between evaluation measures It would be interesting to evaluate whether the introduction of graph-based recommender models led to noticeable changes regarding the relationships between different recommendation objectives and the levels of prediction instability. As graph-based models, and especially GNN-based models, are the

current state-of-the-art in recommendation, it would be interesting to see how these models might have changed the nature of recommendation challenges.

Evaluating the influence of pre-processing on evaluation measures The recommendation objectives evaluated in this thesis are often caused by data-inherent biases. It might be of interest to analyse how different pre-processing approaches influence the distinctiveness of these biases and the correlations between the different recommendation objectives. A possible application could be the oversampling of ratings w.r.t. the different user groups to remove the data-inherent rating imbalance responsible for user unfairness in recommendation results.

Evaluating the influence of post-processing optimisation Similarly to the approach of debiasing during pre-processing, it would be possible to analyse the influence of post-processing approaches on the development of recommendation objectives and their relationships to each other. This would allow for the comparison of the differences between debiasing approaches applied during pre-, in- and post-processing w.r.t. their influence on relationships among recommendation objectives.

Evaluating item biases for different user groups While we have evaluated the overall existence of diversity and popularity bias in recommender algorithms, it might be of interest to combine the evaluation of these item biases with the definition of user fairness. Analysing how these biases might differ for different user groups could offer additional dimensions of user unfairness strongly connected to the overall research interest.

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A.

Experimental setup

A.1. Choice of datasets

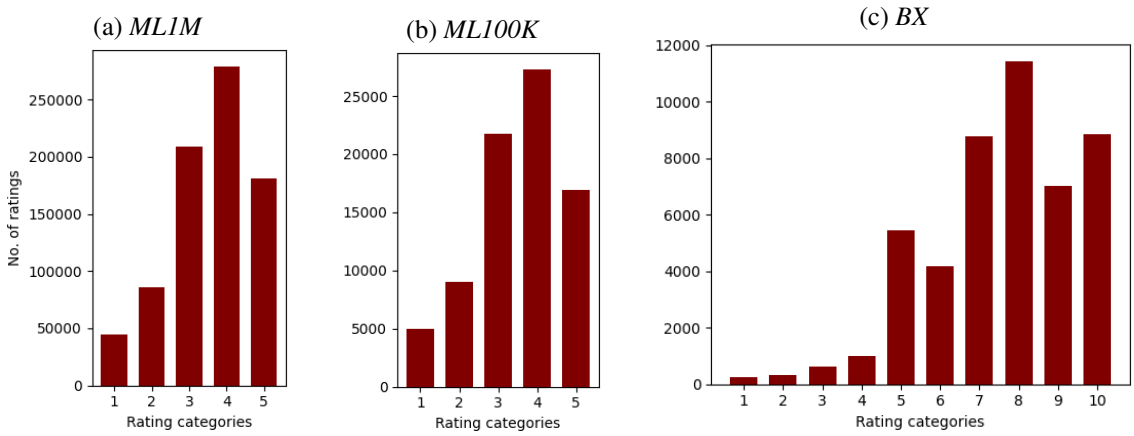

Figure A.1.: ***Distribution of ratings over the different rating categories in the training data.***

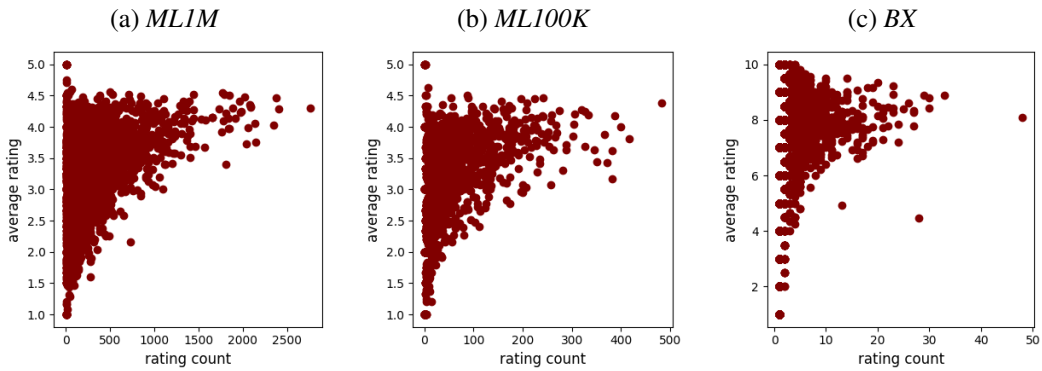

Figure A.2.: ***Correlation between item popularity and average rating in the training data.***

B.

Results

B.1. RQ3 - Correlation analysis

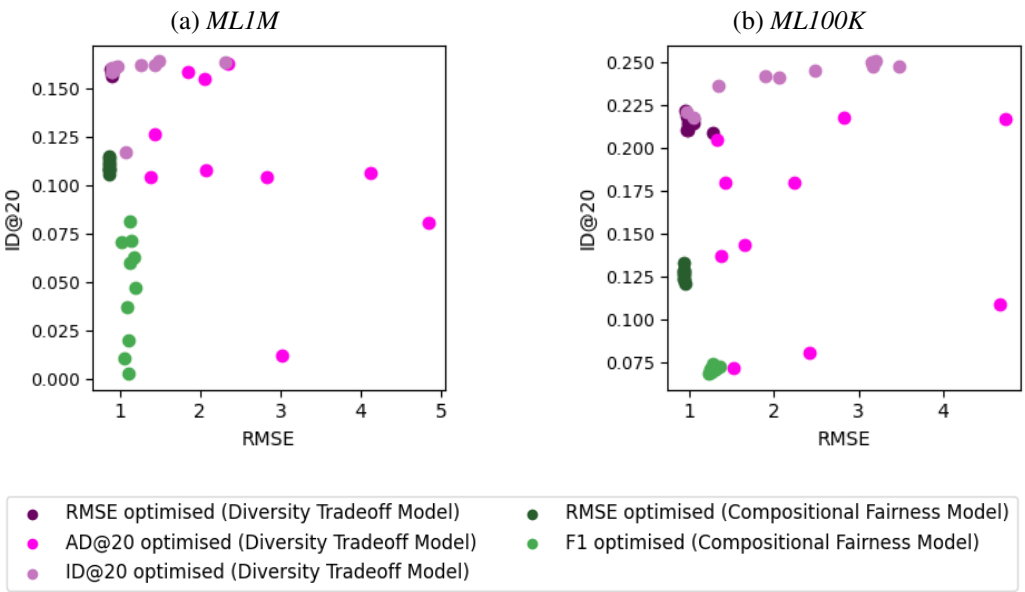

Figure B.1.: ***Scatter plots for prediction performance and individual diversity over all optimisation objectives.***

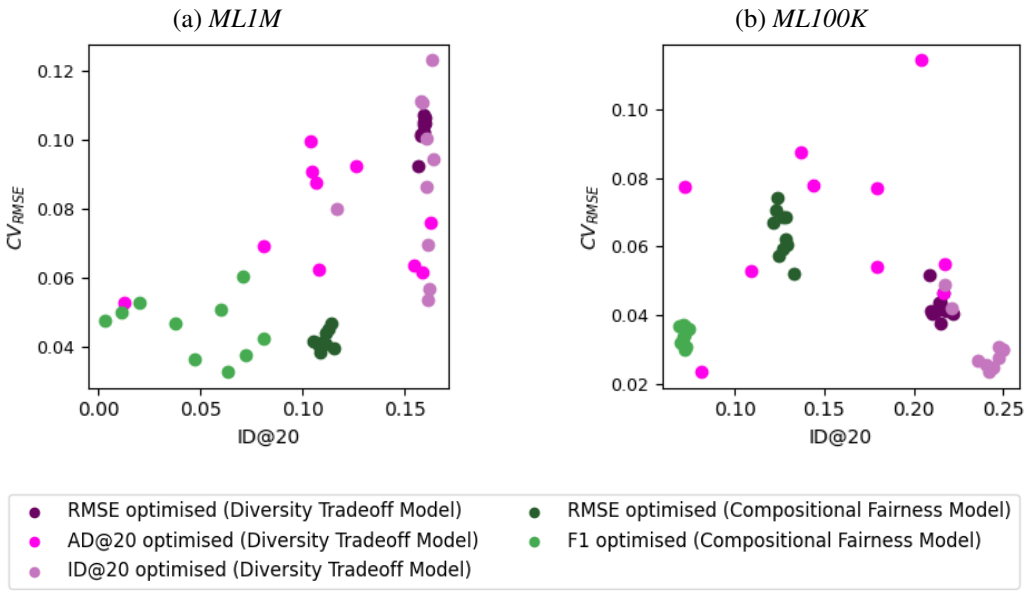

Figure B.2.: ***Scatter plots for individual diversity and user unfairness (CV_{RMSE}) over all optimisation objectives.***

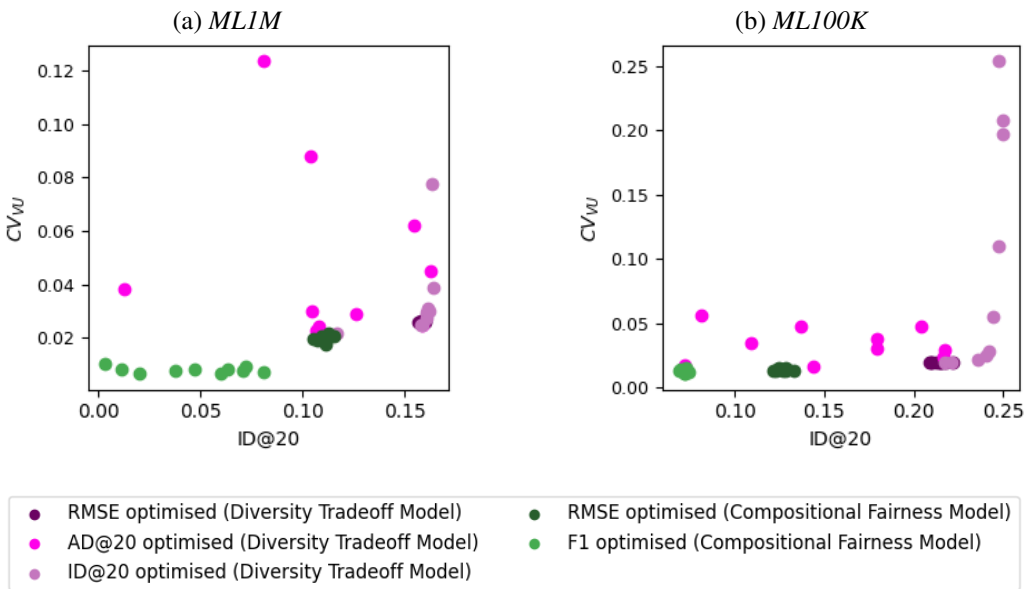

Figure B.3.: ***Scatter plots for individual diversity and user unfairness (CV_{VU}) over all optimisation objectives.***

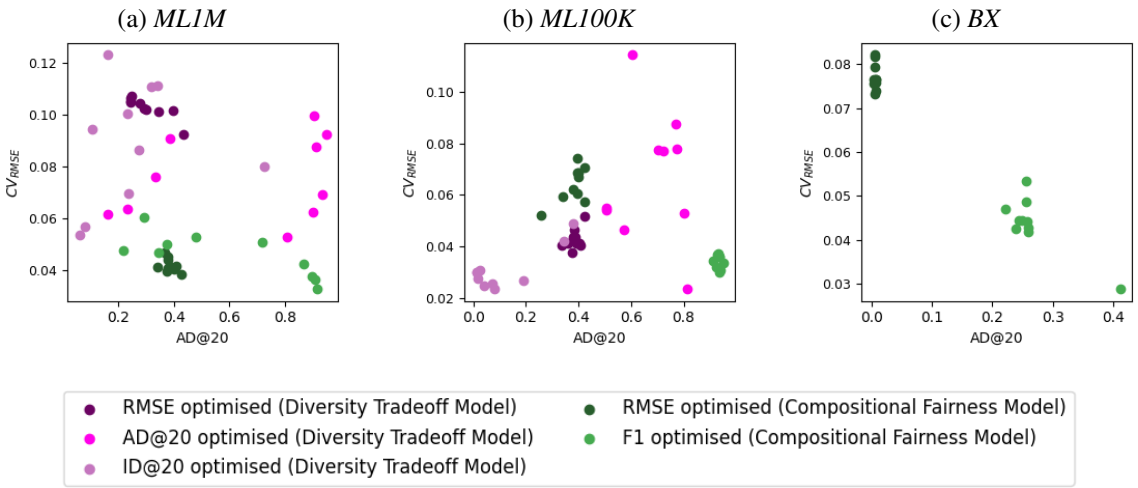

Figure B.4.: ***Scatter plots for aggregated diversity and user unfairness (CV_{RMSE}) over all optimisation objectives.***

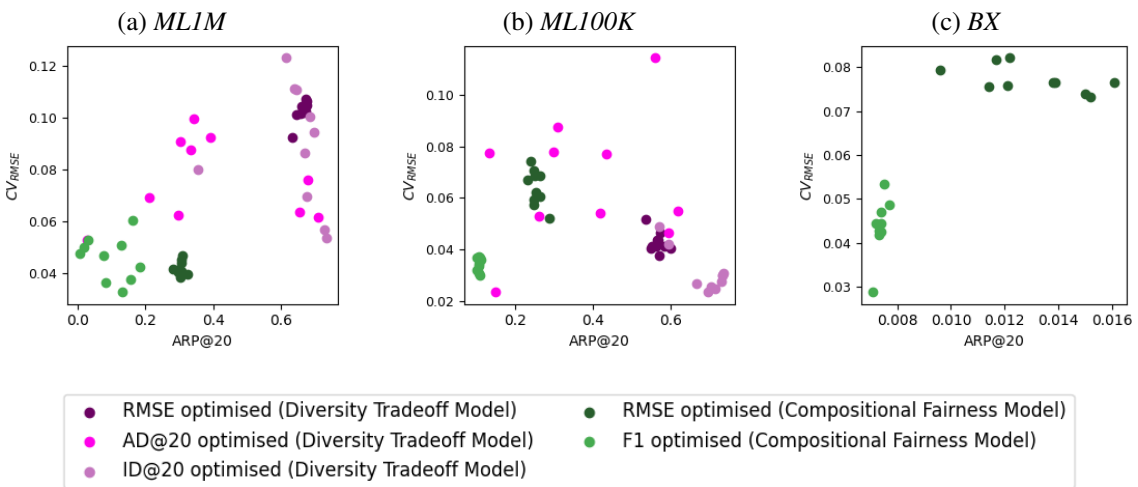

Figure B.5.: ***Scatter plots for popularity bias and user unfairness (CV_{RMSE}) over all optimisation objectives.***

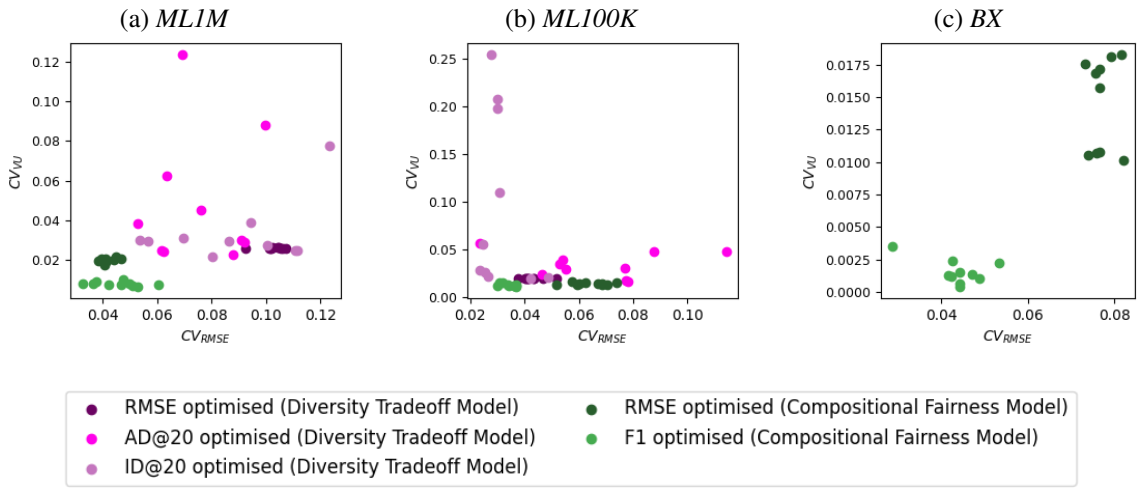

Figure B.6.: ***Scatter plots for user unfairness measured by CV_{RMSE} and CV_{VU} over all optimisation objectives.***

B.2. Case study A - Understanding the definition of fairness for the Compositional Fairness mode

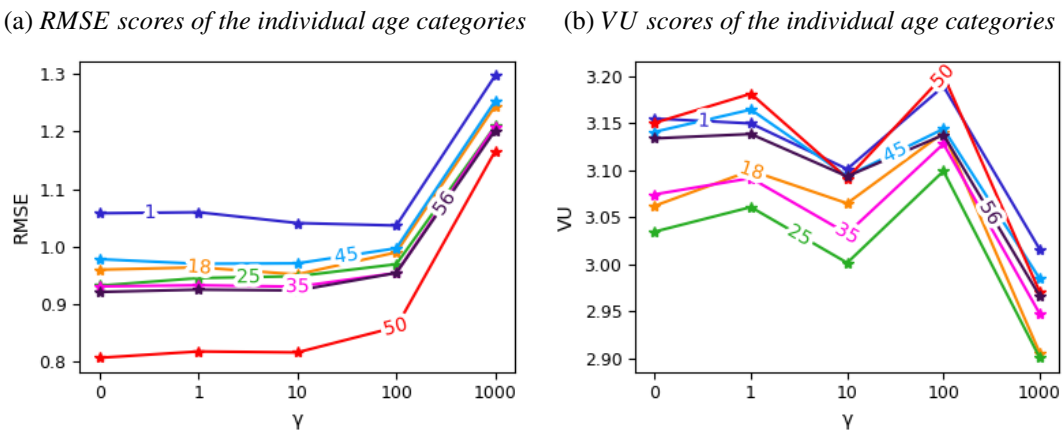

Figure B.7.: ***The influence of varying the fairness objective on the Compositional Fairness model on the RMSE and VU scores of the individual age categories.*** *The results have been produced using the ML100K dataset and are based on one instantiation optimised for prediction performance, and only changing the γ setting when training further instantiations.*

(a) *RMSE scores of the individual age categories* (b) *VU scores of the individual age categories*

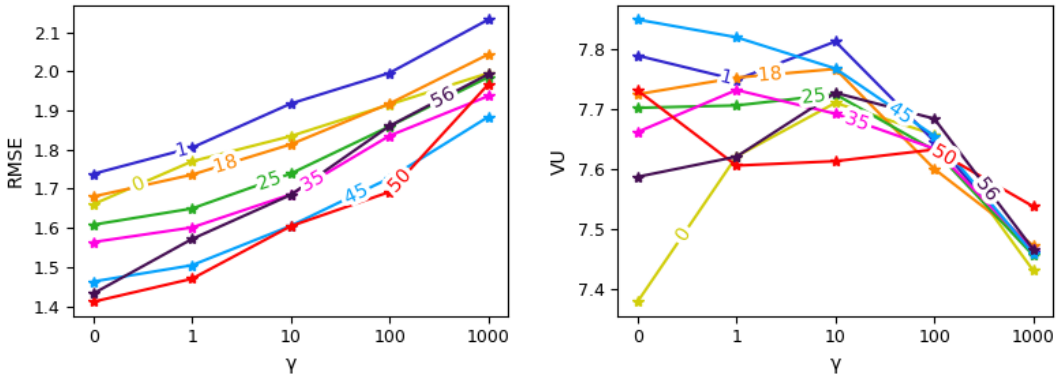

Figure B.8.: ***The influence of varying the fairness objective on the Compositional Fairness model on the** RMSE **and** VU **scores of the individual age categories.*** *The results have been produced using the BX dataset and are based on one instantiation optimised for prediction performance, only changing the γ setting when training further instantiations.*

B.3. Case study B - Influence of the fairness penalty on the performance of the Compositional Fairness model

(a) $F1$

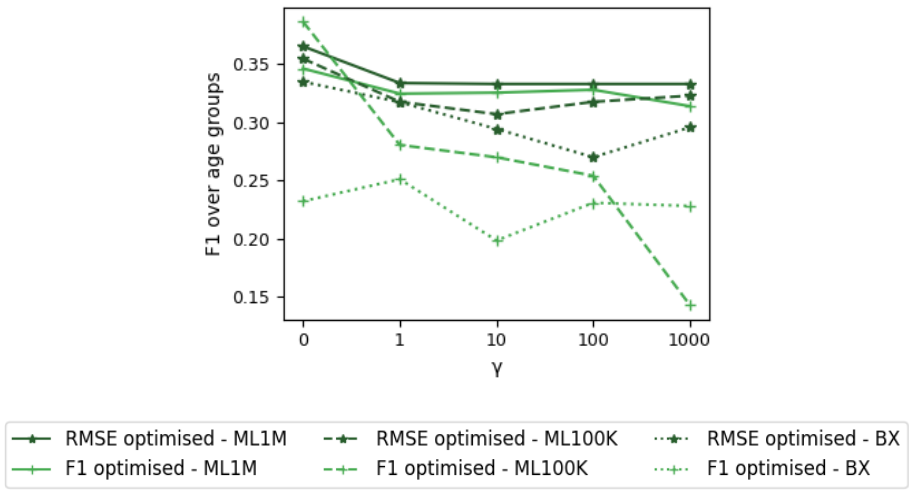

Figure B.9.: ***The influence of varying the fairness objective on the Compositional Fairness model on user fairness measured by the** $F1$ **score.***

B.4. Case study C - Influence of the Furthest Neighbour graph on the performance of the Diversity Tradeoff model

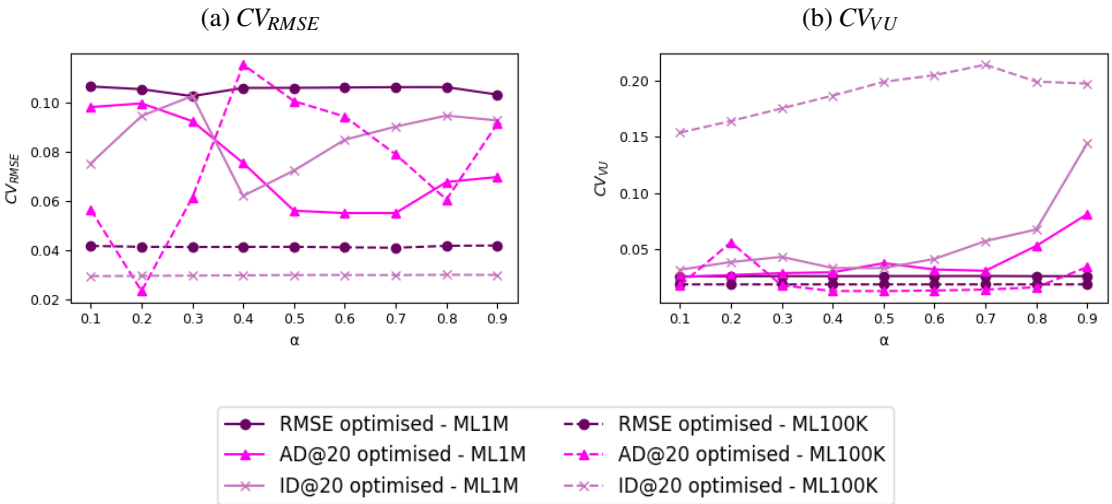

Figure B.10.: ***The influence of varying the impact of the FN graph in the Diversity Tradeoff model on aggregated and individual diversity.***